CTU training solutions | Private Tertiary Education

CAPSTONE PROJECT

FINAL DISSERTATION

Project Title	CampusBook: a lecturer consultation booking system for CTU Training Solutions
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Module	Software Engineering 2 (SPM622)
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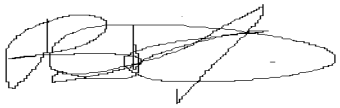
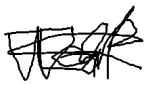
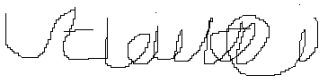
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Declaration of Originality

We the members of SE2 Group 3 declare that this dissertation CampusBook a lecturer consultation booking system for CTU Training Solutions is our own work Where the work of others has been used it has been acknowledged and referenced in the text and in the reference list This work has not been submitted for any other module or qualification

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Abstract

Students at CTU Training Solutions arrange consultations with lecturers through Teams messages WhatsApp requests in class and walk-ins so nobody can see when a lecturer is free and meetings are missed or double-booked A survey of 13 students confirmed the problem 8 could not tell when a lecturer was free 10 waited one to two days for a confirmed time and 7 had given up on getting a consultation

This project designed developed and evaluated CampusBook a responsive web application where lecturers publish the times they are free and students book change and cancel consultations with a short reason for the visit The system uses a React front end a Node.js and Express API and a MySQL database Confirmation and reminder emails are sent with Nodemailer and a database transaction with a row lock and a unique key stops two students booking the same slot

The team worked in one-week Scrum sprints and used GitHub branches and pull requests 36 test cases were linked to the requirements and all 10 defects found in testing were fixed and retested All 9 functional requirements were met the slowest page loaded in 389 ms and eight booking requests sent for one slot at the same moment produced one booking In user testing with 4 students and 2 lecturers 4 of the 6 users rated CampusBook easy to use The students averaged 4.5 out of 5 and the lecturers 2.5 which shows that the lecturer page is the main area to improve

Keywords: consultation booking, web application, lecturer availability, scheduling, Scrum

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List of Abbreviations and Acronyms

Abbreviation	Meaning
API	Application Programming Interface
CSS	Cascading Style Sheets
CTU	CTU Training Solutions
ERD	Entity Relationship Diagram
FK	Foreign Key
FR	Functional Requirement
HTML	HyperText Markup Language
HTTP / HTTPS	HyperText Transfer Protocol (Secure)
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
ms	Milliseconds
NFR	Non-Functional Requirement
PK	Primary Key
POPIA	Protection of Personal Information Act
PR	Pull Request
REST	Representational State Transfer
RTM	Requirements Traceability Matrix
SMS	Short Message Service
SMTP	Simple Mail Transfer Protocol
SQL	Structured Query Language
SRS	Software Requirements Specification
TC	Test Case
UML	Unified Modelling Language
US	User Story

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Chapter 1 Introduction

1.1 Background

Consultations give students time with a lecturer outside class to ask questions get feedback and work through problems Research shows this support is often under used and that the reasons are usually practical rather than academic (Griffin, 2014)

At CTU Training Solutions arranging a consultation is informal Students send an email a Teams or WhatsApp message or simply knock on the lecturer door during posted hours Each lecturer handles this differently and nobody has a clear view of who is available who is coming or which meetings took place

Other sectors have solved a similar problem with online booking in healthcare web based appointment systems have reduced no-shows and waiting times and saved staff time (Zhao, 2017) and automated reminders improve attendance even further (Guy, 2012) With 96% of South African households owning a mobile phone (Statistics South Africa, 2024) a booking system that works well on a phone can reach almost every student

1.2 Problem Statement

CTU Training Solutions has no single reliable way for students to book consultations with lecturers Requests are spread across email Teams WhatsApp and walk-ins so students wait outside offices messages go unanswered and meetings are missed or double-booked Lecturers cannot plan their time because they do not know who is coming or why the problem peaks before tests and assignment deadlines exactly when students need help most

It affects students who need academic support and lecturers who juggle requests from several channels Our group has experienced this first hand our lecturer reviewed the proposal and agreed it addresses a real need and a short student survey showed how widespread it is

1.3 Aim

To design develop and evaluate CampusBook a web based system that lets CTU students book change and cancel consultations and lets lecturers manage their availability so that fewer consultations are missed or left unplanned

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1.4 Objectives

1. Analyse how consultations are arranged today through a survey of at least 10 students by 21 October 2026
2. Compare at least three existing scheduling systems on their features strengths and limitations by 19 October 2026
3. Design and develop a web system with all six must-have features working on a live database by 23 October 2026
4. Test the system so that every must-have test case passes a booking takes under 2 minutes and pages load within 3 seconds
5. Evaluate the system with at least 5 students and 1 lecturer with 80% or more rating it easy to use

1.5 Research Questions

1. How do students and lecturers at CTU arrange consultations now and what problems do they face?
2. What do existing scheduling systems offer and what do they lack for a campus setting?
3. How can a web system let students book consultations and lecturers manage their time?
4. Does the system meet its functional and non-functional requirements?
5. Do students and lecturers find the system easy to use and useful?

Proposed Software Solution

CampusBook is a responsive web application with two roles Lecturers publish their weekly consultation slots and see who has booked and why Students browse open slots book one with a short reason for the visit and get an email reminder before the meeting The six must have features are:

1. Login with separate student and lecturer roles
2. Lecturers publish weekly consultation slots
3. Students book change or cancel a slot with a reason for the visit
4. Email confirmation and a reminder before each booking
5. Booking history for students and lecturers

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6. Mobile friendly design

Scope

Table 1.1 *Scope of CampusBook*

In scope	Out of scope
One CTU campus	Other campuses or institutions
Student and lecturer roles	Admin reporting dashboards
Six must have features with email reminders	Video calls SMS and native mobile apps
Test data created by the team	Links to CTU live student systems

Significance of the Study

- Students can see when lecturers are free get confirmed bookings and stop wasting trips before assessments
- Lecturers control their consultation time and know in advance who is coming and why
- CTU Training Solutions gains a low-cost tool that supports student success and evidence on whether online booking works as well on campus as it does in healthcare

Chapter Summary

This chapter explained why consultations at CTU Training Solutions are hard to arrange and who the problem affects It set the aim the five objectives and the five research questions and described CampusBook with its six must have features and what is in and out of scope Chapter 2 now reviews existing research and similar systems

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Chapter 2 Literature Review

2.1 Introduction

This chapter reviews research on why students under use consultations and how online booking and reminders help in other sectors It then compares Microsoft Bookings Calendly and Moodle Scheduler and reviews the technology chosen for CampusBook before stating the research gap

2.2 Review of Related Literature

Consultations are under used Griffin (2014) surveyed undergraduate students at a large American university and found that most of the factors that stop students attending office hours are outside the lecturer control The one exception was feedback: students who found it useful came back more often A summary of similar survey work found that 66% of more than 600 students had never attended office hours for a course and only 8% went more than once a month (Weimer, 2015) Both sources show the problem clearly but they focus on student attitudes rather than on how meetings are arranged and neither tests a tool that makes booking easier

Online booking works in healthcare Zhao (2017) reviewed 21 web based medical appointment systems Online booking cut no shows with one system reporting 2% for web bookings against 8% for phone bookings and it also shortened waiting times raised satisfaction and reduced staff workload The review warns that providers worry about losing control of their schedules a concern lecturers are likely to share Guy (2012) add that automated reminders increase attendance These studies come from healthcare where appointments are more formal than a quick consultation so the idea still needs testing in education

People use systems that are useful and easy The Technology Acceptance Model suggests that people adopt a system when they find it useful and easy to use (Davis, 1989) Nielsen (1994) usability heuristics turn this into practical design rules such as always showing a clear confirmation and stopping a student from booking a slot that is already taken

2.3 Review of Similar Systems

Microsoft Bookings offers a booking page that syncs with Outlook sends email reminders and creates Teams meetings It comes with Microsoft 365 business and education plans but SMS reminders only work in North America and the UK and need an extra Teams Premium licence (Microsoft, 2025) It is built

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for businesses serving customers so it has no idea of students lecturers or modules and setting it up needs administrator access

Calendly is a popular polished scheduling tool Its free plan allows only one event type and one calendar and reminders need the Standard plan at 10 US dollars per lecturer per month (Calendly, 2025) Students would need a separate link for every lecturer and booking data would sit with a third party

Moodle Scheduler is a Moodle plugin that lets teachers create slots for students to book with reminders booking limits and attendance tracking (Moodle, 2017) It suits teaching well but it only works inside one course needs an administrator to install it and is only an option for institutions that run Moodle

Table 2.1 *Existing systems compared with CampusBook*

	Microsoft Bookings	**Calendly**	**Moodle Scheduler**	**CampusBook**
Lecturer sets slots	Yes	Yes	Yes	Yes
Student books and cancels	Yes	Yes	Yes	Yes
Free reminders	Email only	No paid plan	Yes	Yes email
Student and lecturer roles	No	No	Course roles	Yes
All lecturers in one place	Yes	No	No per course	Yes
Extra cost	Microsoft 365 licence	Per lecturer	Needs Moodle	None
Set up effort	Admin sets up a booking page	Each lecturer signs up	Admin installs it then added per course	Set up once for the campus
Where data is kept	Microsoft 365 tenant	Calendly servers	Moodle server	CTU server
Works on a phone	Yes in the browser	Yes browser and app	Through Moodle	Yes built for phones first
Main drawback	Business focus	Cost and data privacy	Tied to one course	Must be maintained

Lessons for CampusBook All three systems handle the basic act of booking well so CampusBook copies what works: clear open slots instant confirmation and reminders Where they fall short is fit Microsoft Bookings and Calendly are designed for businesses and Calendly charges per lecturer for reminders Moodle Scheduler fits teaching but stays inside a single course CTU current mix of email Teams WhatsApp and walk ins costs nothing but offers no shared view of availability and no record of meetings

2.4 Review of Technologies

The chosen tools are among the most widely used in the 2025 Stack Overflow Developer Survey so documentation and help are easy to find (Stack Overflow, 2025)

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CampusBook stores names email addresses and booking details which counts as personal information under the Protection of Personal Information Act (Republic of South Africa, 2013) The system collects only what it needs hashes all passwords shows each user only their own bookings and used made up data during development

Table 2.2 *Technology choices for CampusBook*

Part	Choice	Why	Alternative
Front end	React	Builds pages from reusable components Meta used by about 45% of developers	Plain HTML and JavaScript harder to keep consistent
Back end	Node.js and Express	One language JavaScript across the whole team used by about 49% of developers	ASP.NET which adds a second language
Database	MySQL	Relational tables for users slots and bookings and the team already knows it	PostgreSQL more popular at about 56% but less familiar to us
Reminders	Email	Free and reaches everyone	SMS which costs per message
Method	Scrum	Sprints of a month or less with a review after each (Schwaber and Sutherland, 2020) which fits our weekly deadlines	Waterfall too rigid for weekly feedback

2.5 Research Gap

Research shows that consultations help students but are under used (Griffin, 2014) and that online booking with reminders reduces missed appointments in healthcare (Guy, 2012; Zhao, 2017) Existing tools serve businesses or single Moodle courses well but none gives CTU students one place to book any lecturer free of licence costs

Conceptual and Technical Direction

Two ideas guide the design From the Technology Acceptance Model every screen must make booking quicker than sending a message or students and lecturers will not switch (Davis, 1989) From the healthcare studies confirmations and reminders are what reduce missed meetings so they are core features rather than extras (Guy, 2012; Zhao, 2017) Lecturers keep full control of their own slots which answers the loss of control concern raised by Zhao (2017)

Technically CampusBook follows a simple three-tier design The React front end runs in the browser the Express API handles logins slots and bookings and the MySQL database stores the data with rules that stop a slot being booked twice A scheduled task sends reminder emails before each consultation

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2.6 Conclusion

Research shows that consultations help students but are under used and that online booking with reminders reduces missed appointments in healthcare The three systems reviewed handle booking well but none gives CTU students one place to book any lecturer free of licence costs Chapter 3 turns these ideas into requirements and a method

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Chapter 3 Requirements and Methodology

3.1 Introduction

This chapter sets out who will use CampusBook what it must do and how the team built it It covers the stakeholders the functional and non-functional requirements how the requirements were gathered the Scrum method the user stories the architecture and the ethical privacy and security considerations

3.2 Stakeholder Identification

Table 3.1 *Stakeholders and what they need*

Stakeholder	What they need from CampusBook	Access
Students	See open consultation times book change or cancel a slot and get a reminder before the meeting	Student account
Lecturers	Publish their available times and see who is coming and why	Lecturer account
CTU Training Solutions	Better planned consultations and more support for students before assessments	No admin dashboard in this version
Project supervisor	Review the requirements ethics and each weekly submission	Review only
Development team	Design build and test the system using made-up data	Development environment

3.3 Requirements Gathering

The first requirements came from three sources Our own experience of booking consultations at CTU feedback from our lecturer on the proposal the research comparison of Microsoft Bookings Calendly and Moodle Scheduler The features the three systems share became core requirements and the gaps they leave such as one place for every lecturer and free reminders shaped what makes CampusBook different

An anonymous survey of 13 CTU students checked how students book consultations now and what problems they run into and the results confirmed these requirements (Chapter 5)

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3.4 Functional Requirements

Table 3.2 *Functional requirements*

ID	Requirement	Priority
FR-01	Students and lecturers log in with their email and password	Must
FR-02	The system checks the user role so students and lecturers only reach their own pages and actions	Must
FR-03	Lecturers publish edit and remove their own future consultation slots	Must
FR-04	Students view the open slots of a chosen lecturer	Must
FR-05	Students book an open slot and give a short reason for the visit	Must
FR-06	Students change or cancel their own upcoming bookings	Must
FR-07	The system stops two students booking the same slot	Must
FR-08	The system sends an email confirmation after booking and a reminder before the meeting	Must
FR-09	Students see their own booking history and lecturers see the bookings for their slots	Must

3.5 Non-Functional Requirements

Table 3.3 *Non-functional requirements*

ID	Requirement	Priority
NFR-01	A first-time student completes a booking in under 2 minutes	Must
NFR-02	The main pages load within 3 seconds in the team test set up	Must
NFR-03	Every page works on a phone at 375 pixels wide and on a laptop at 1366 pixels wide	Must
NFR-04	Passwords are stored as bcrypt hashes and never as plain text	Must
NFR-05	Each user can only see their own bookings or the bookings for their own slots	Must
NFR-06	Only the data needed for logging in and booking is stored with made-up data during development	Must
NFR-07	Errors explain what to fix and the form keeps what the user already typed	Should

3.6 Use Cases and User Stories

Table 3.4 *User stories*

ID	User story	Requirements
US-01	As a student I want to see open slots so I can get help without messaging each lecturer	FR-04
US-02	As a student I want to book a slot with a reason so the lecturer knows what I need	FR-05 and FR-07
US-03	As a student I want to change or cancel my booking so my plans stay up to date	FR-06
US-04	As a lecturer I want to publish and manage my slots so I control my own time	FR-03
US-05	As a lecturer I want to see who booked and why so I can prepare for the meeting	FR-09
US-06	As a student or lecturer I want a confirmation and a reminder so no meeting is forgotten	FR-08

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The main use case is booking a slot a student logs in chooses a lecturer and an open slot adds a reason and confirms the system checks that the slot is still open saves the booking and sends the confirmation If another student took the slot first the student is asked to choose another time

3.7 Methodology (Agile Scrum)

The team uses Scrum with one week sprints where each sprint ends with a review so the next one can improve on it (Schwaber and Sutherland, 2020)

Table 3.5 *Sprint plan*

Sprint	Focus	Ends with
Sprint 0 16 to 19 Oct	Requirements SRS system design wireframes and repository set up	Week 3 submission
Sprint 1 20 to 23 Oct	Build the six must-have features and run the first tests	Week 4 submission
Sprint 2 24 to 29 Oct	User testing bug fixes final dissertation and presentation	Final submission and live demo

Table 3.6 *Scrum compared with Waterfall*

	Scrum	Waterfall
Feedback	Used every week	Mostly at the end
Changing requirements	Added to the next sprint	Hard once a phase is done
Risk	Problems show up early	Problems show up late in testing
Fit for this project	Matches the weekly deadlines	Too rigid for a two-week build

Planned System Architecture

CampusBook uses a three tier design The React front end runs in the browser the Express API handles logins slots and bookings and the MySQL database stores the data with rules that stop a slot being booked twice A scheduled task sends reminder emails before each consultation The full architecture diagram is in Chapter 4 section 4.2

3.8 Ethics and Privacy

- Only made-up users and bookings are used during development never real CTU records
- The survey and user testing are voluntary with consent first and anonymous results as set out in Appendix B
- Research files are kept in a private folder and deleted once the project is marked
- Only the data needed for logging in and booking is collected in line with POPIA (Republic of South Africa, 2013)

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3.9 Security

Security was designed in from the start and checked again in testing (Chapter 5)

- Passwords are hashed every request is checked for a valid token and role and all input is validated on the server
- A unique key in the database stops a slot ever having two confirmed bookings
- After 10 wrong passwords in 15 minutes the login refuses more tries for a while and an expired session sends the user back to the login page

Chapter Summary

This chapter listed the stakeholders the functional and non-functional requirements and the Scrum method the team follows It also set out how privacy and security are handled Chapter 4 turns these requirements into the system design

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Chapter 4 System Design, Implementation and Version Control

4.1 Introduction

This chapter shows how CampusBook is designed It covers the architecture the UML diagrams the database design and the user interface wireframes followed by the technology stack and the repository structure

4.2 System Architecture

CampusBook uses a three tier architecture The client tier is a React web app that runs in the browser on a phone or laptop The application tier is a Node.js and Express API that holds all business rules The data tier is a MySQL database Splitting the system this way keeps the interface the rules and the data separate so each team member can work on one tier without breaking the others

Every request from the browser goes to the API over HTTPS as JSON Middleware checks the login token and the user role and validates the input before any route runs the auth slot and booking routes then read and write the database A scheduler runs every minute finds bookings that are due a reminder and sends them through the email service Booking confirmations are sent straight after a booking is saved

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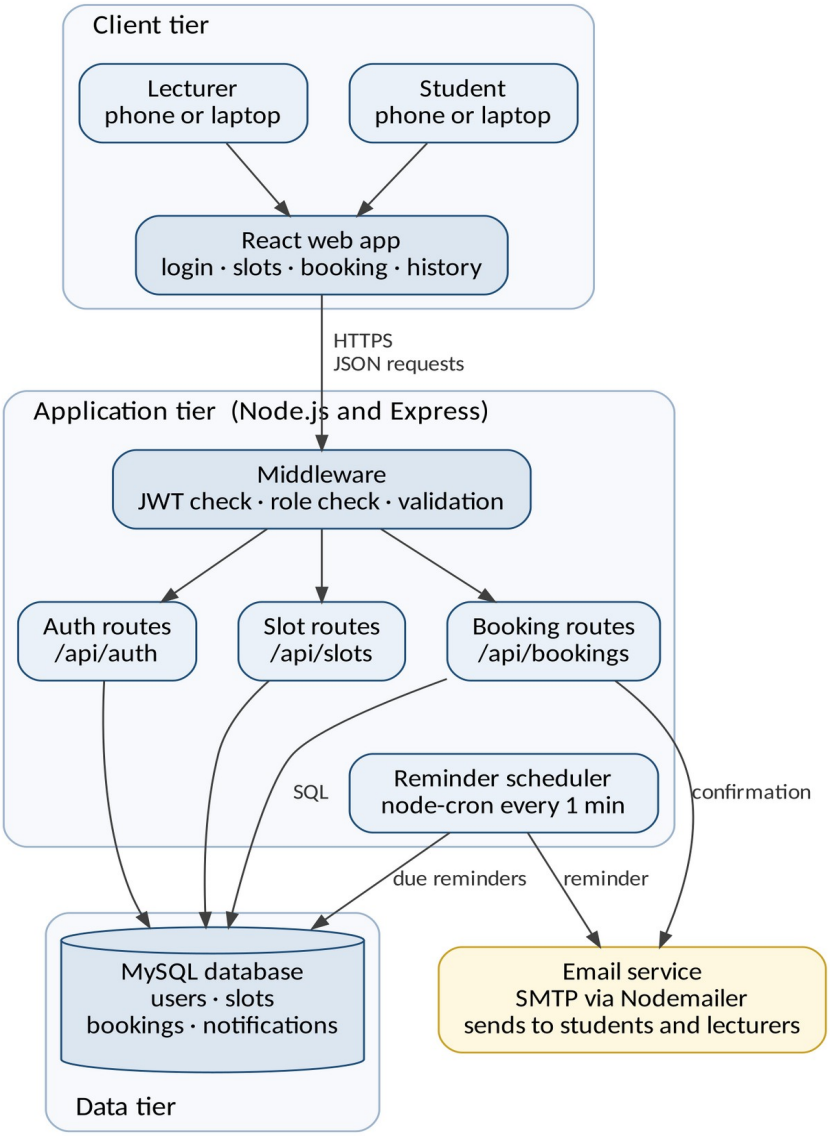


Figure 4.1 *CampusBook three tier architecture*

4.3 UML Design

Use case diagram

The use case diagram shows what each user can do Students view open slots book change and cancel bookings Lecturers publish edit and remove slots and see upcoming bookings Both log in and view their

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booking history Booking a slot always includes sending a confirmation and reminder through the email service

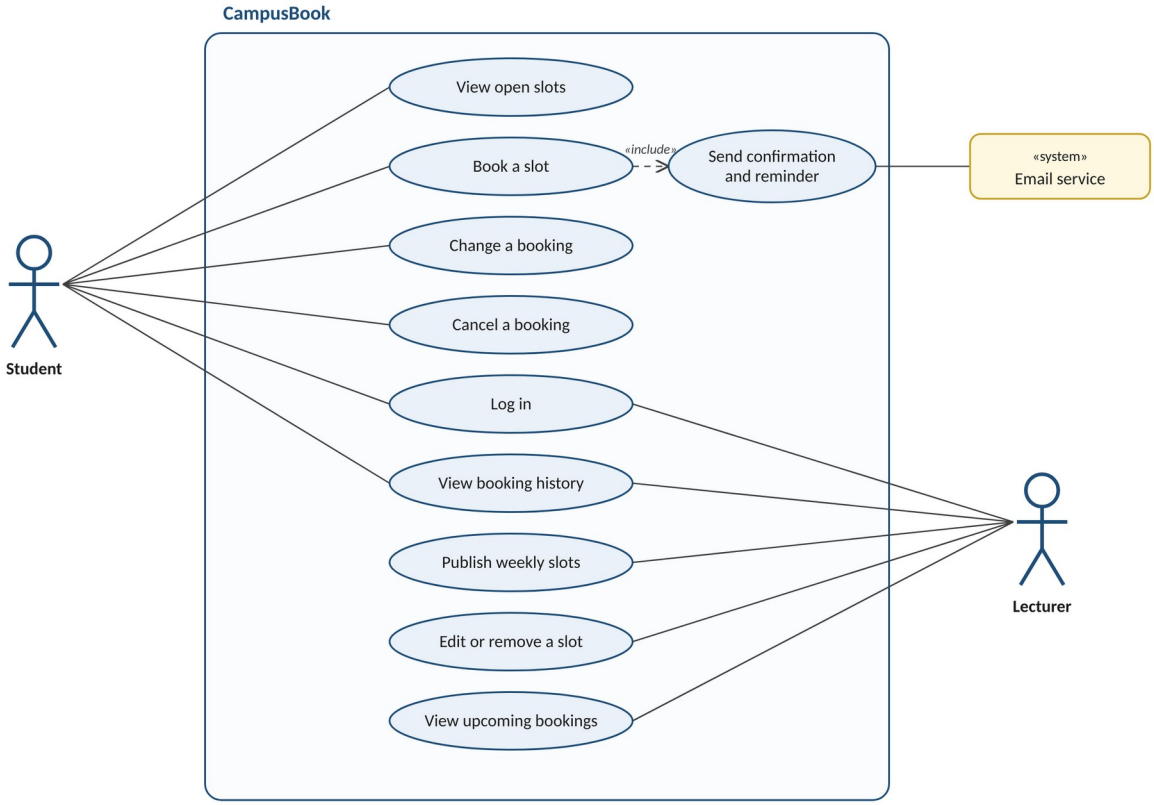


Figure 4.2 *Use case diagram*

Class diagram

The class diagram shows the main objects in the system Student and Lecturer both inherit from User which holds the login details A lecturer publishes many consultation slots and a slot holds at most one active booking Each booking triggers one or more notifications such as the confirmation and the reminder

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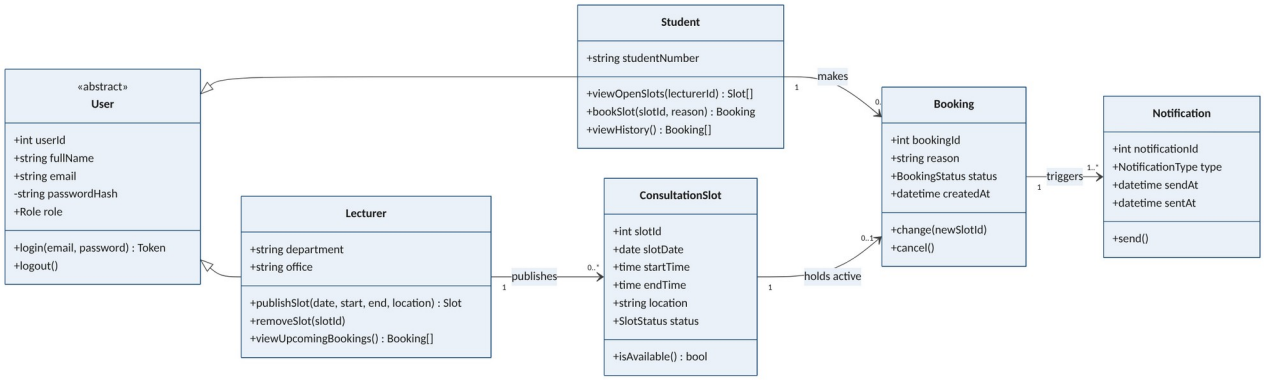


Figure 4.3 *Class diagram*

Activity diagram

The activity diagram follows a student booking a consultation It shows the two key decisions If no suitable slot is open the student can check another week or lecturer If another student takes the slot first the system shows a message and refreshes the list instead of creating a double booking

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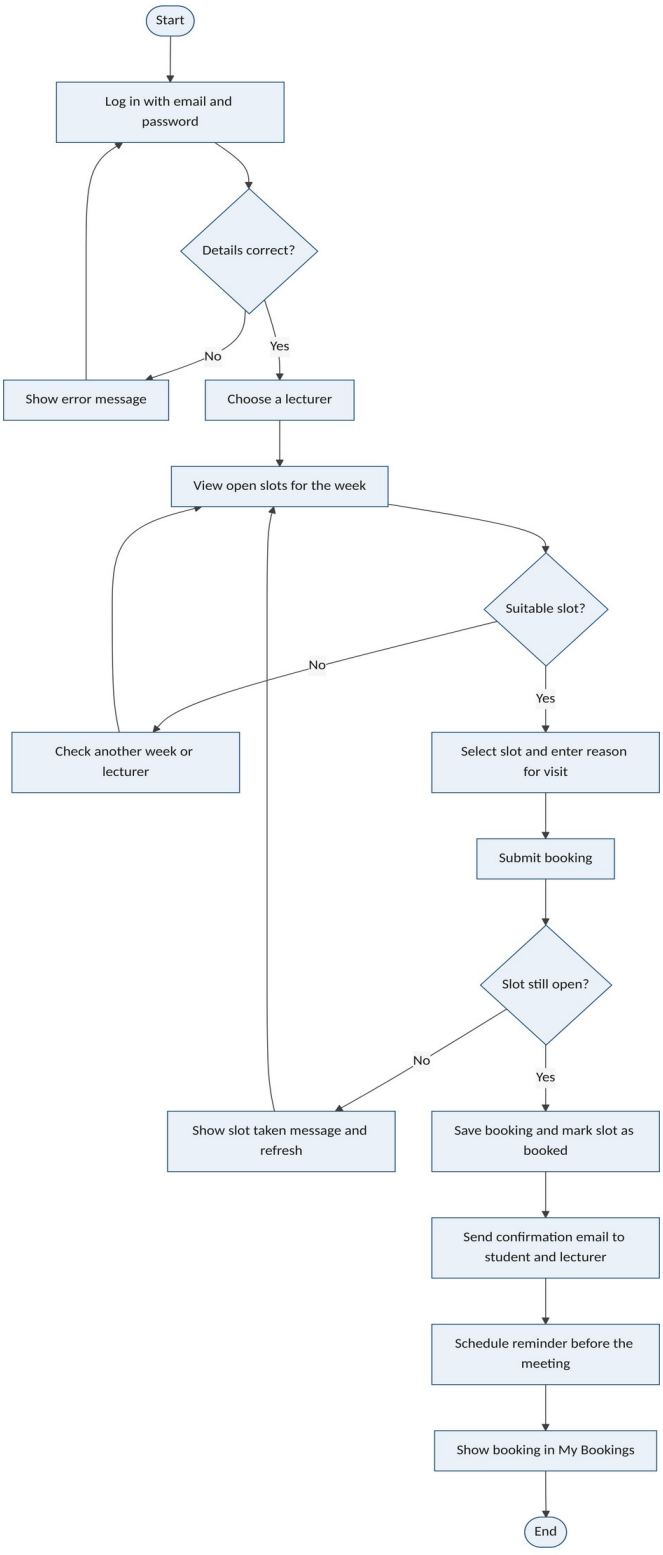


Figure 4.4 *Activity diagram for booking a consultation*

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Sequence diagram

The sequence diagram shows the messages behind a booking The API locks the slot row inside a database transaction so two students pressing book at the same moment cannot both succeed If the slot is open the booking is saved the confirmation email is sent and the student sees a confirmation If not the transaction is rolled back and the student is told the slot was taken

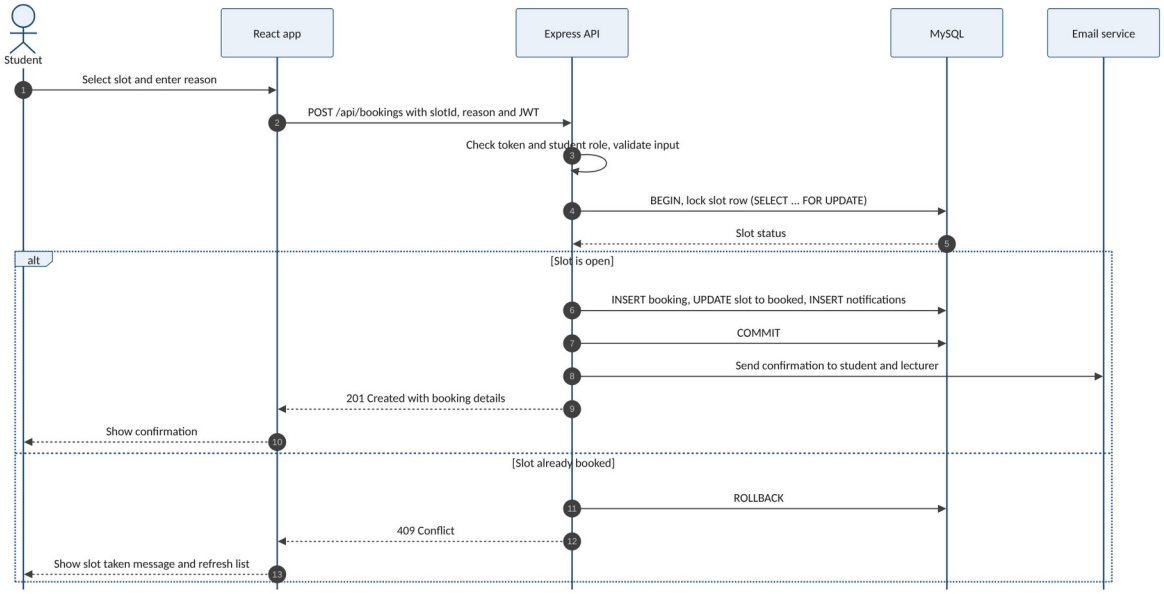


Figure 4.5 *Sequence diagram for booking a slot*

4.4 Database Design

The database stores users lecturer details consultation slots bookings and notifications The ERD below shows the five tables their keys and how they relate

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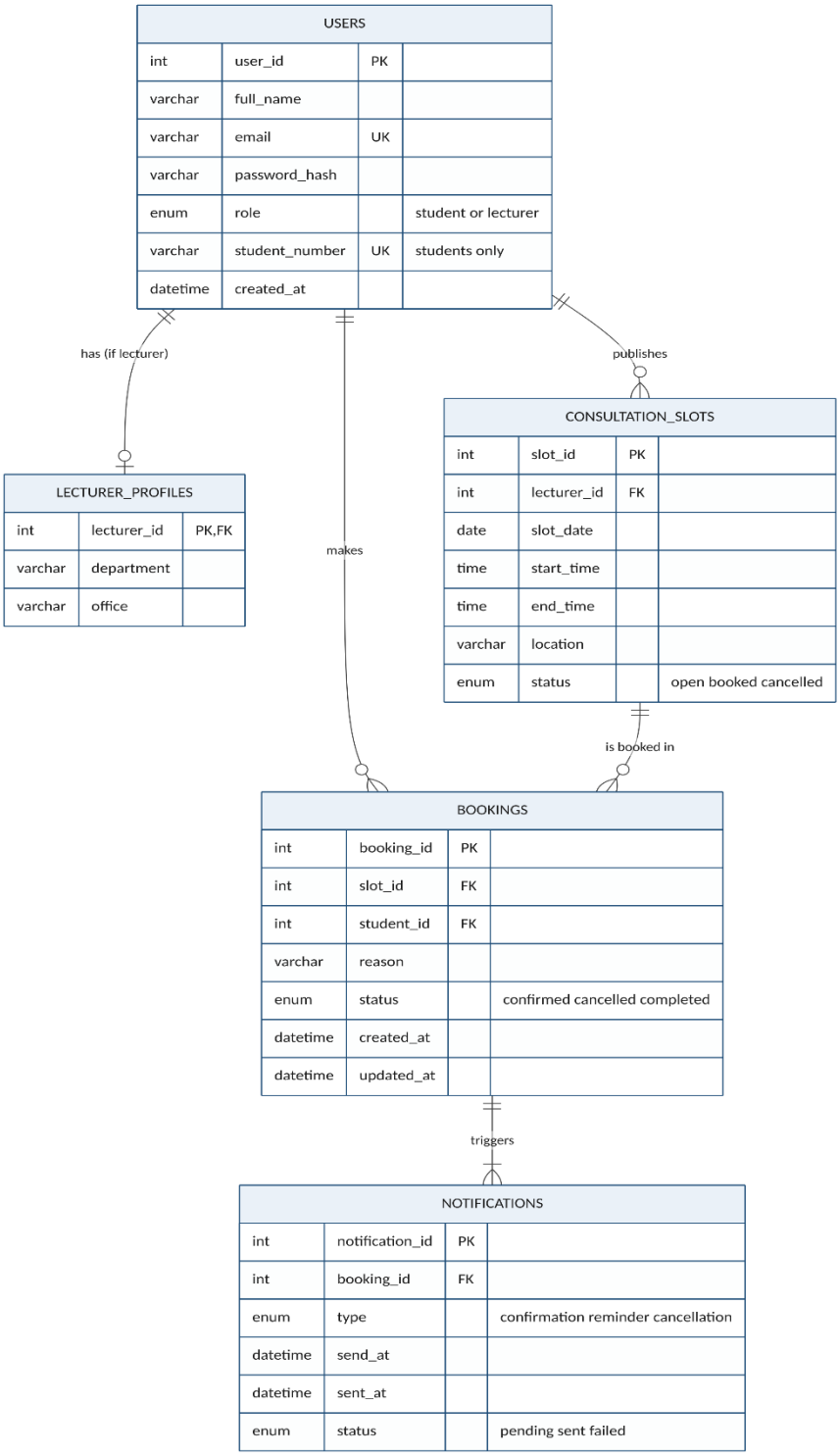


Figure 4.6 *Entity relationship diagram*

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Relationships and cardinality

Table 4.1 *Relationships and cardinality*

Relationship	Cardinality	Meaning
users to lecturer_profiles	One to zero or one	Only lecturers have a profile
users to consultation_slots	One to many	A lecturer publishes many slots
users to bookings	One to many	A student makes many bookings
consultation_slots to bookings	One to many	A slot can be booked again after a cancellation but only one booking is active at a time
bookings to notifications	One to one or more	Each booking has a confirmation and a reminder

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Data dictionary

Table 4.2 *Data dictionary*

Column	Type	Null	Key	Description
users				
user_id	INT UNSIGNED	No	PK	Unique number for each user
full_name	VARCHAR(120)	No		Name shown in the app
email	VARCHAR(150)	No	Unique	Used to log in and receive emails
password_hash	VARCHAR(255)	No		bcrypt hash never the real password
role	ENUM	No		student or lecturer
student_number	VARCHAR(20)	Yes	Unique	Required for students only
lecturer_profiles				
lecturer_id	INT UNSIGNED	No	PK and FK	Links to users
department	VARCHAR(100)	No		Department the lecturer belongs to
office	VARCHAR(50)	Yes		Office or room
consultation_slots				
slot_id	INT UNSIGNED	No	PK	Unique number for each slot
lecturer_id	INT UNSIGNED	No	FK	Lecturer who owns the slot
slot_date	DATE	No		Day of the consultation
start_time and end_time	TIME	No		End must be after start
location	VARCHAR(100)	No		Room or Teams
status	ENUM	No		open booked or cancelled
bookings				
booking_id	INT UNSIGNED	No	PK	Unique number for each booking
slot_id	INT UNSIGNED	No	FK	Slot that was booked
student_id	INT UNSIGNED	No	FK	Student who booked
reason	VARCHAR(255)	No		Reason for the visit
status	ENUM	No		confirmed cancelled or completed
active_slot_id	INT UNSIGNED	Yes	Unique	Filled only while confirmed stops double booking
notifications				
notification_id	INT UNSIGNED	No	PK	Unique number for each email
booking_id	INT UNSIGNED	No	FK	Booking the email belongs to
type	ENUM	No		confirmation reminder or cancellation
send_at and sent_at	DATETIME	sent_at only		When it is due and when it was sent
status	ENUM	No		pending sent or failed

Normalisation

The design is in third normal form Every table has a single column primary key and every other column depends only on that key Lecturer details sit in their own table so they are not repeated for students Lecturer and student names are stored once in users and joined when needed so changing a name

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never leaves old copies behind Notifications are kept apart from bookings so that one booking can have several emails without repeating booking data

Security and integrity

- Foreign keys stop a booking pointing at a slot or student that does not exist
- A unique key on active_slot_id means a slot can only ever have one confirmed booking
- Check constraints make sure end times are after start times and that students have a student number
- Passwords are stored as bcrypt hashes and each user can only reach their own bookings through the API
- Only made-up data is used during development in line with POPIA

SQL scripts

The full scripts are in the repository at database/schema.sql and database/seed.sql The part that prevents double booking is shown below

```
CREATE TABLE bookings (
    booking_id      INT UNSIGNED NOT NULL AUTO_INCREMENT,
    slot_id         INT UNSIGNED NOT NULL,
    student_id      INT UNSIGNED NOT NULL,
    reason          VARCHAR(255) NOT NULL,
    status          ENUM('confirmed','cancelled','completed') DEFAULT 'confirmed',
    active_slot_id  INT UNSIGNED AS (IF(status = 'confirmed', slot_id, NULL)) STORED,
    PRIMARY KEY (booking_id),
    UNIQUE KEY uq_one_active_booking_per_slot (active_slot_id),
    CONSTRAINT fk_booking_slot FOREIGN KEY (slot_id) REFERENCES consultation_slots (slot_id),
    CONSTRAINT fk_booking_student FOREIGN KEY (student_id) REFERENCES users (user_id)
);
```

4.5 User Interface Design

The screens below are low fidelity wireframes that show the layout and flow of each page before it was built in React in Week 4 The numbered markers point to the design choices explained under each screen The layout follows the usability heuristics from Chapter 2 (Nielsen, 1994) and keeps every task to as few steps as possible so that booking is quicker than sending a message (Davis, 1989)

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Login

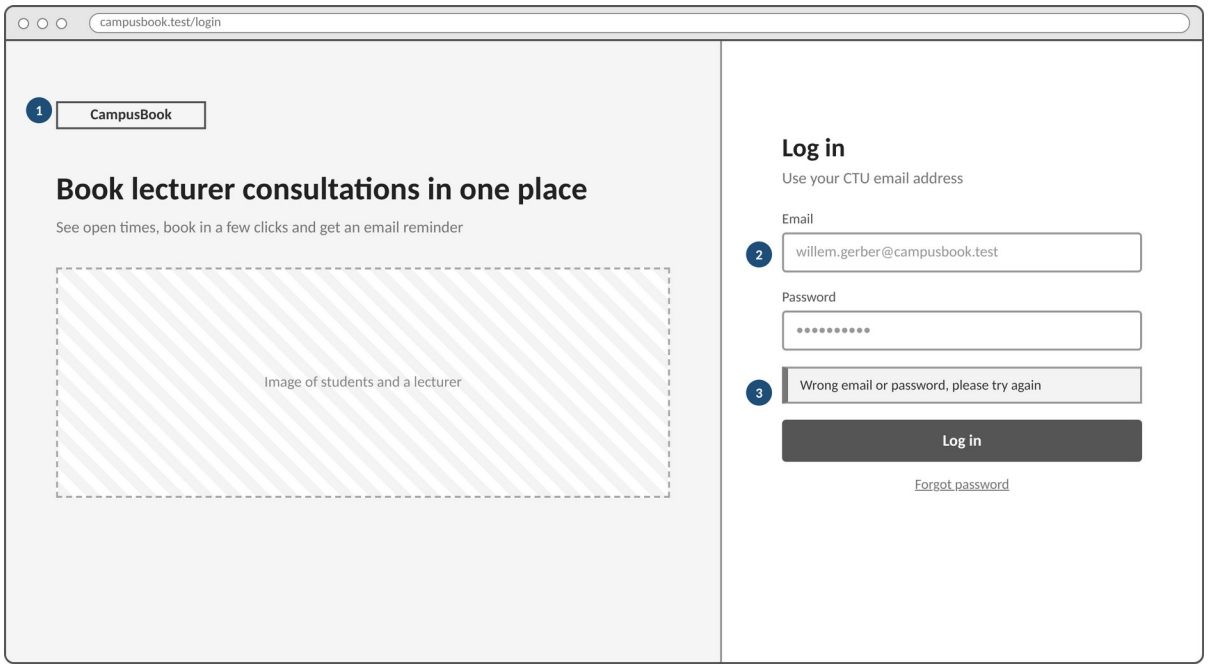


Figure 4.7 *Login wireframe*

1. The left side explains in one line what CampusBook does so new users know they are in the right place
2. Labels sit above each field and the email field hints that a CTU address is needed
3. Errors appear in plain words right above the button so users can recognise and recover from a mistake

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Student booking screen

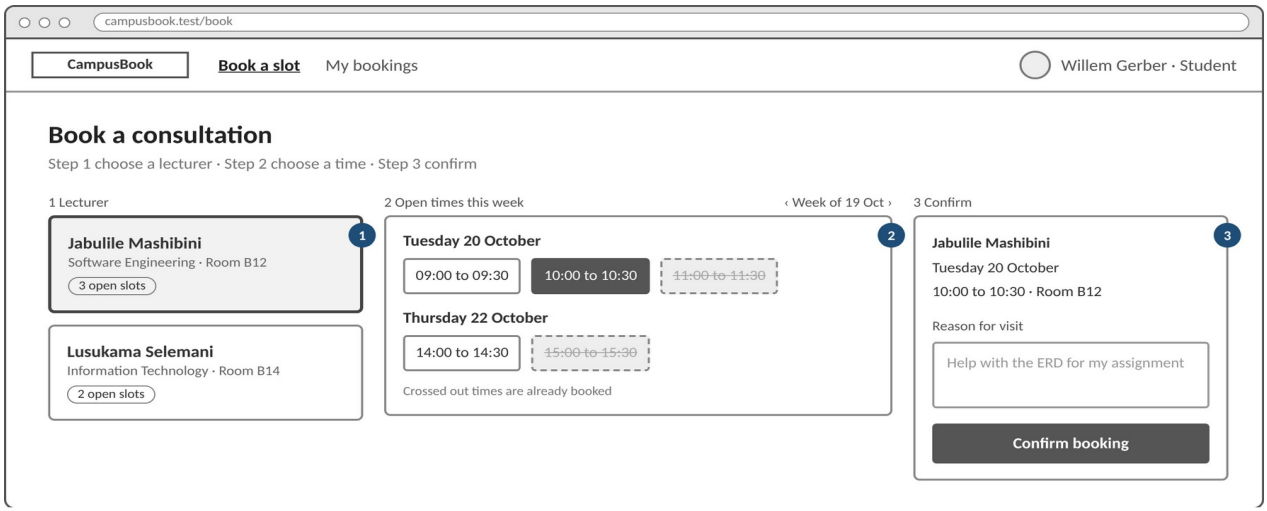


Figure 4.8 *Student booking screen wireframe*

1. Lecturer cards show the department room and number of open slots so students can choose quickly
2. Open times are grouped by day and times that are already taken are crossed out instead of hidden which prevents errors
3. A summary repeats the lecturer date time and place before the student confirms and the reason for the visit is required

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Lecturer slots screen

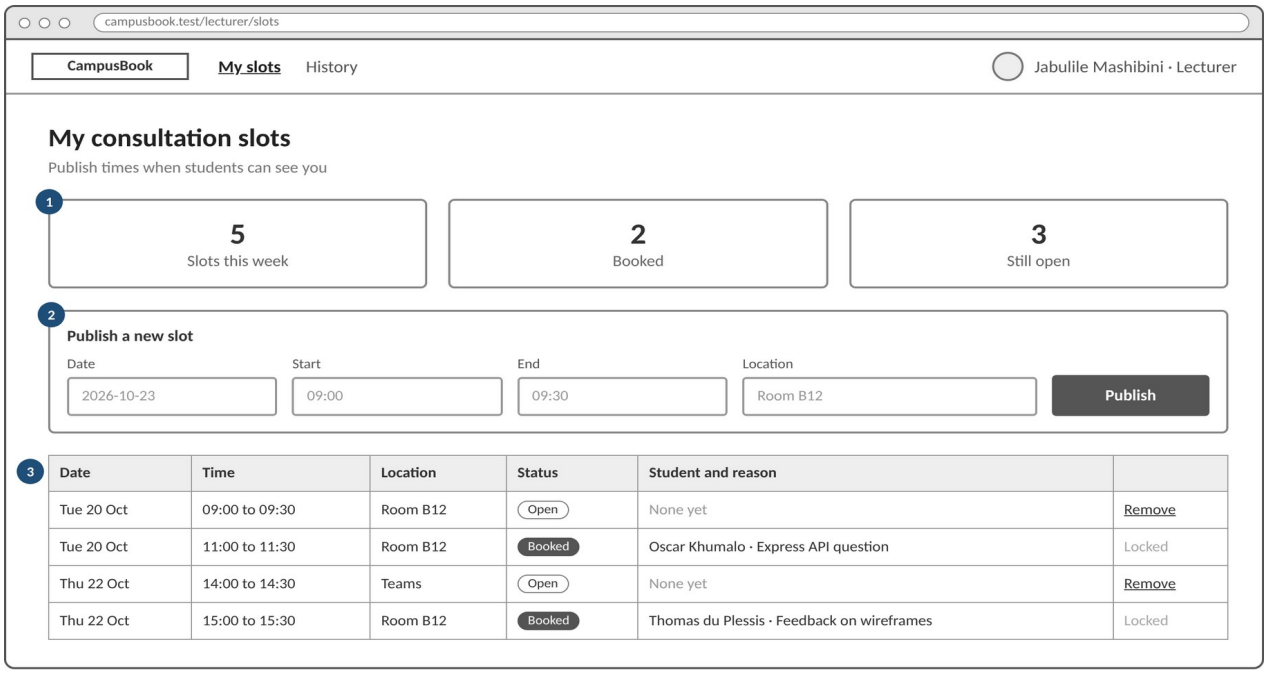


Figure 4.9 *Lecturer slots screen wireframe*

1. Three counters show the week at a glance so lecturers always know the status of the system
2. A new slot is published with four fields on one line which keeps the most common lecturer task short
3. The table lists every slot with the student and reason and booked slots are locked so a confirmed meeting cannot be removed by accident

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My bookings on a phone

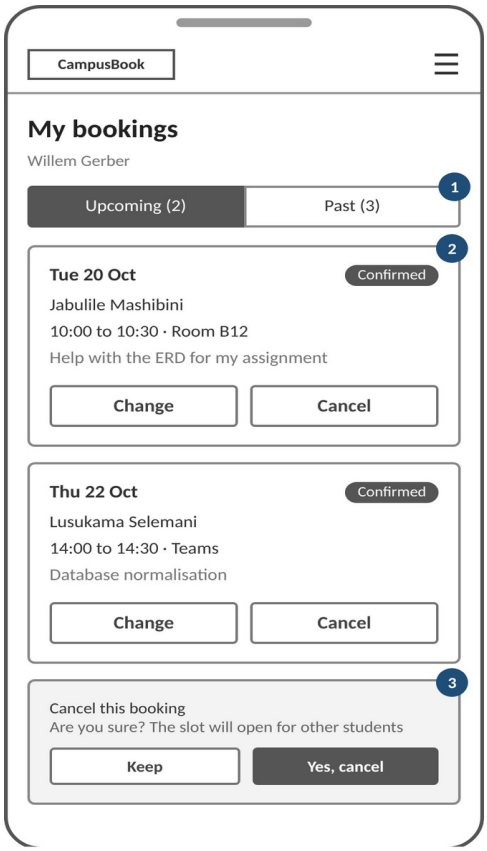


Figure 4.10 *My bookings wireframe on a phone*

1. Upcoming and past bookings sit in two tabs so the list stays short on a small screen
2. Each card shows the date lecturer time place and reason with large buttons that are easy to tap
3. Cancelling asks for confirmation first and explains that the slot will open for other students

4.6 Implementation

CampusBook was built with a small set of tools that suit a secure and simple consultation booking system

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Table 4.3 *Development tools*

Area	Tools	What they do
Languages	JavaScript HTML CSS SQL	JavaScript runs the frontend and the backend HTML gives the page structure CSS controls the look and the small screen layout SQL creates and reads the database tables
Frontend	React Vite React Router	React gives reusable parts for login booking and lecturer slots Vite is the fast dev server React Router moves between pages and blocks pages by role
Backend	Node.js Express	Node.js runs JavaScript on the server Express handles the REST API for login slots and bookings and talks to the database
IDE	Visual Studio Code	Used to write and organise the code The built in terminal installs packages runs both servers and runs Git commands
Database	MySQL 8	Stores accounts slots bookings and notifications Keys and constraints link the records and stop double bookings
Libraries	bcryptjs jsonwebtoken Nodemailer node-cron	bcryptjs hashes passwords JSON Web Tokens handle login and roles Nodemailer sends emails node-cron runs the reminder job
Services	SMTP GitHub	SMTP sends booking confirmations and reminders Both servers run locally for testing GitHub stores and shares the source code
Version control	Git GitHub	Each member works on their own branch Changes are committed pushed and merged through pull requests This shows who did what and keeps the latest CampusBook version

4.7 Working System

CampusBook is a web-based lecturer consultation booking system developed using React for the frontend, Node.js and Express for the backend, and MySQL for data storage. It provides separate functions for students and lecturers: students can find available consultation times and manage bookings, while lecturers can publish and manage their availability. The modules communicate through a REST API and work together to support the main consultation-booking workflow.

Authentication and Login Module

The authentication module allows students and lecturers to access the system using their login credentials. The application distinguishes between the two user roles so that each user can access the functionality relevant to them. The backend uses authentication and role-based middleware to protect restricted API endpoints. The booking routes require an authenticated user, while specific operations are restricted to students or lecturers.

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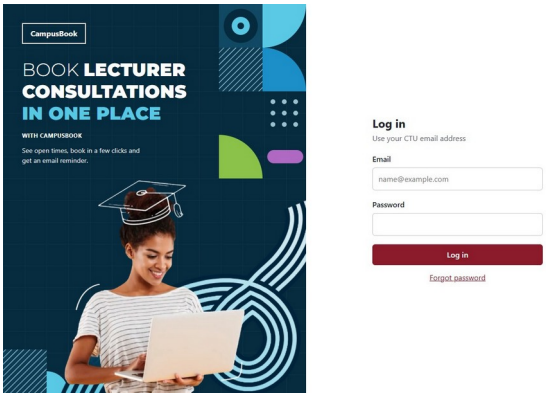


Figure 4.11 *CampusBook login page*

Code snippet: Protected booking routes

```
const express = require('express');
const pool = require('../db');
const { requireAuth, requireRole } = require('../middleware/auth');
const { isId, isText, validateBody } = require('../middleware/validate');
const mailer = require('../services/mailer');

const router = express.Router();
router.use(requireAuth);

const REMINDER_MINUTES = Number(process.env.REMINDER_MINUTES_BEFORE || 60);
```

This code applies the authentication middleware to the booking routes. It helps ensure that booking operations are not accessible to unauthenticated users.

Consultation Slot Management Module

The consultation slot management module enables lecturers to publish consultation times and manage their availability. Students can view available slots and select a suitable time for a consultation.

The backend stores slot information in the consultation_slots table. Each slot is associated with a lecturer, date, start time, end time, location and status. The slot status indicates whether the slot is open, booked or cancelled. This module supports the main objective of providing a centralised method for displaying and managing lecturer consultation availability.

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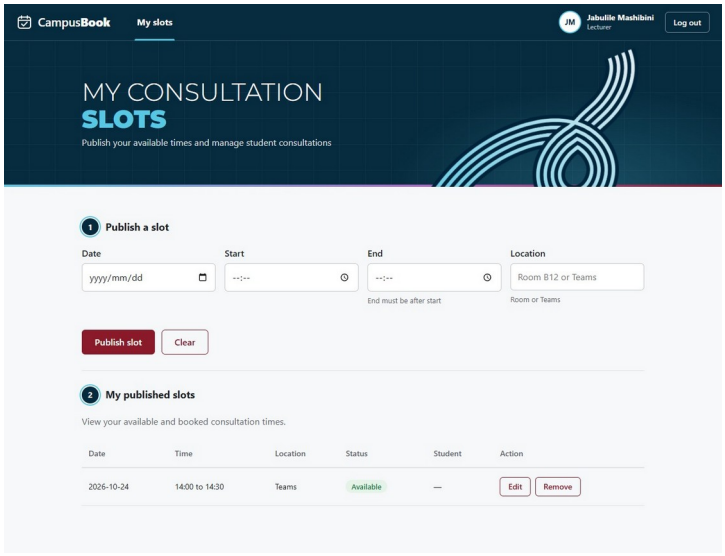


Figure 4.12 *Lecturer slots page*

Code snippet: Slot time and overlap validation (backend/src/routes/slots.js)

```
async function checkSlotTime(db, lecturerId, slot, ignoreId = 0) {
  if (slot.end <= slot.start) throw httpError(400, 'The end time must be after the start time');

  const [[{ isFuture }]] = await db.query('SELECT TIMESTAMP(?, ?) > NOW() AS isFuture', [slot.date,
slot.start]);
  if (!isFuture) throw httpError(400, 'Choose a time in the future');

  const [overlap] = await db.query(
    `SELECT slot_id FROM consultation_slots
     WHERE lecturer_id = ? AND slot_date = ? AND status <> 'cancelled' AND slot_id <> ?
       AND start_time < ? AND end_time > ?
     LIMIT 1`,
    [lecturerId, slot.date, ignoreId, slot.end, slot.start]
  );
  if (overlap.length) throw httpError(409, 'This overlaps another one of your slots');
}
```

Student Booking Module

The student booking module allows a student to select an available consultation slot and provide a short reason for the visit. The backend checks that the slot exists, is still open and starts in the future. The booking operation runs inside a database transaction and locks the slot row while it is processed, helping prevent two students from booking the same time. A database uniqueness constraint also protects against duplicate bookings.

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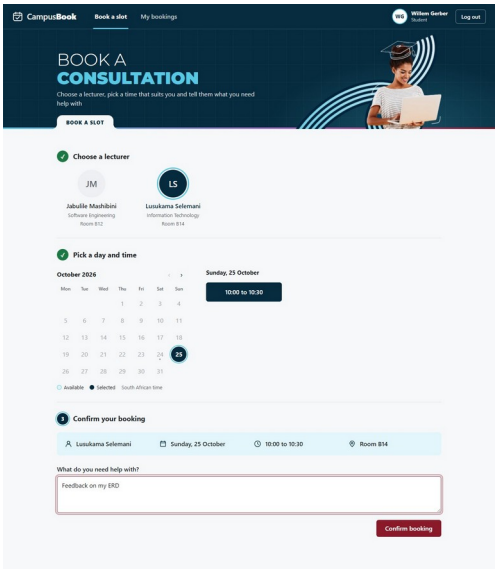


Figure 4.13 *Student booking page*

Code snippet: Booking availability validation

```
async function bookSlot(conn, slotId, studentId, reason) {
  const slot = await lockSlot(conn, slotId);
  if (!slot || slot.status !== 'open' || !slot.isFuture) throw httpError(409, SLOT_TAKEN);

  const [result] = await conn
    .query('INSERT INTO bookings (slot_id, student_id, reason) VALUES (?, ?, ?)', [slotId, studentId,
reason])
    .catch(err => { throw err.code === 'ER_DUP_ENTRY' ? httpError(409, SLOT_TAKEN) : err; });

  await conn.query("UPDATE consultation_slots SET status = 'booked' WHERE slot_id = ?", [slotId]);
  await conn.query(
    `INSERT INTO notifications (booking_id, type, send_at) VALUES
       (?, 'confirmation', NOW()),
       (?, 'reminder', TIMESTAMP(?, ?) - INTERVAL ? MINUTE)`,
    [result.insertId, result.insertId, slot.date, slot.start, REMINDER_MINUTES]
  );

  return { bookingId: result.insertId, slot };
}
```

This code verifies that the requested slot exists, is open and has not already passed. If these conditions are not satisfied, the booking is rejected. Otherwise it saves the booking, marks the slot as booked and creates the confirmation and reminder notifications.

Booking History and Management Module

The booking management module allows students to view their bookings. Lecturers can also retrieve bookings associated with their consultation slots. The backend provides separate endpoints for student and lecturer booking information. These functions help users keep track of scheduled consultations and manage changes when necessary.

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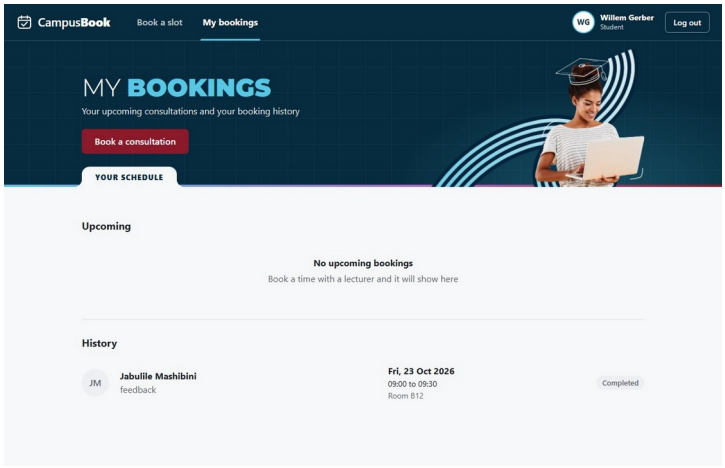


Figure 4.14 *Student My bookings page*

Code snippet: Retrieving student bookings

```
router.get('/mine', requireRole('student'), async (req, res, next) => {
  try {
    const [rows] = await pool.query(
      `SELECT ${BOOKING_COLUMNS}, u.full_name AS lecturerName
         FROM bookings b
         JOIN consultation_slots s ON s.slot_id = b.slot_id
         JOIN users u ON u.user_id = s.lecturer_id
        WHERE b.student_id = ?
        ORDER BY s.slot_date DESC, s.start_time DESC`,
      [req.user.id]
    );
    res.json(toBookingList(rows));
  } catch (err) {
    next(err);
  }
});
```

The route restricts access to students. It retrieves the booking information from the database and returns the results to the frontend.

Database Connectivity Module

The database module connects the Express backend to MySQL. It provides a reusable connection pool that allows different parts of the backend to execute database queries.

The database contains tables for users, lecturer profiles, consultation slots, bookings and notifications. Relationships between these tables associate students with bookings and lecturers with consultation slots.

Code snippet: Creating the database connection pool

```
const path = require('path');
```


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```
require('dotenv').config({
  path: path.resolve(__dirname, '../.env'),
});

const mysql = require('mysql2/promise');

const pool = mysql.createPool({
  host: process.env.DB_HOST || 'localhost',
  port: Number(process.env.DB_PORT || 3306),
  database: process.env.DB_NAME || 'campusbook',
  user: process.env.DB_USER || 'root',
  password: process.env.DB_PASSWORD || '',

  waitForConnections: true,
  connectionLimit: 10,
  queueLimit: 0,
});

module.exports = pool;
```

The connection settings are read from environment variables rather than being permanently written into the source code. The exported pool is reused by backend routes that need to access MySQL.

The database connection was tested successfully during development.

Email Notification and Reminder Module

The email module uses Nodemailer to send booking-related emails through an SMTP server. These messages are intended to inform students and lecturers about bookings and provide reminders before scheduled consultations.

The system records notification information in the notifications table, including the notification type, scheduled send time, delivery status and time sent. The reminder job uses node-cron to periodically check for reminders that are due.

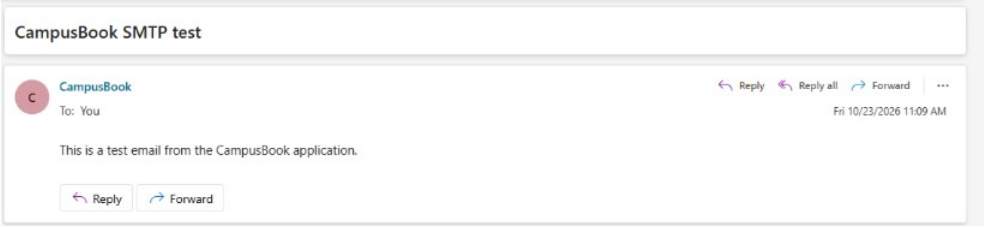


Figure 4.15 *SMTP test email*

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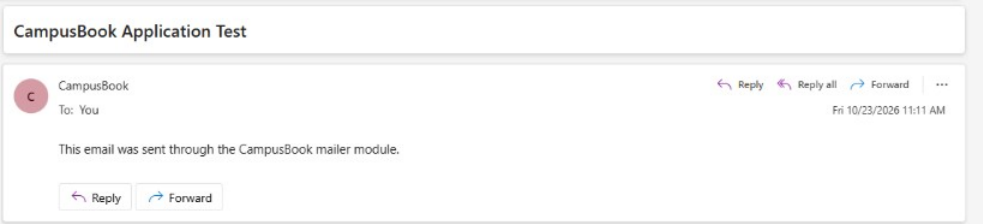


Figure 4.16 *Test email sent through the CampusBook mailer module*

The SMTP connection and the mailer module were tested successfully with a test email.

Code snippet: Sending a booking confirmation

```
async function sendEmail(to, subject, text) {
  try {
    await mailer.sendMail(to, subject, text);
  } catch (err) {
    console.error(`Email to ${to} failed: ${err.message}`);
  }
}

async function emailBoth(user, slot, subject, studentMessage, lecturerMessage) {
  await Promise.all([
    sendEmail(user.email, subject, studentMessage),
    sendEmail(slot.lecturerEmail, subject, lecturerMessage),
  ]);
}

async function markConfirmationSent(bookingId) {
  await pool.query(
    "UPDATE notifications SET status = 'sent', sent_at = NOW() WHERE booking_id = ? AND type =
'confirmation'",
    [bookingId]
  );
}
```

This illustrates how the booking route sends messages to both the student and lecturer and then marks the confirmation as sent in the notifications table. A failed email is logged and does not stop the booking.

User Interface and Responsive Design

The React frontend provides the pages through which students and lecturers interact with CampusBook. The interface is intended to make consultation management accessible through a web browser and to support use on mobile devices.

The pages correspond to the main user activities, including logging in, selecting and booking a consultation slot, managing bookings and publishing lecturer availability.

Summary of the key modules

The key modules work together to support CampusBook's consultation-booking workflow. The React frontend provides role-specific pages, the Express backend validates requests and enforces

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authentication and access control, and MySQL stores user, slot, booking and notification data. The Slots API validates time ranges, future dates and overlaps, while the booking routes protect slot availability, update slot status and provide booking history. Nodemailer and the scheduled reminder job support email communication with students and lecturers.

Implementation Screenshots

This section shows how the main features of CampusBook work in the running system

The system uses React for the frontend with Node.js and Express for the backend and MySQL for users slots and bookings

The screenshots follow the login the student booking flow and the lecturer slot pages and also show errors found during testing and how they were fixed

Repository structure

```
CampusBook/
backend/

Express REST API (routes middleware services jobs db.js server.js)
frontend/
 React web app (pages Login BookSlot MyBookings LecturerSlots)
database/
 schema.sql and seed.sql
docs/


 diagrams and wireframes
```


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Login

Desktop login

The login page on a desktop screen

The left side introduces CampusBook and the right side has the email and password fields

There is a login button and a forgot password link under the form

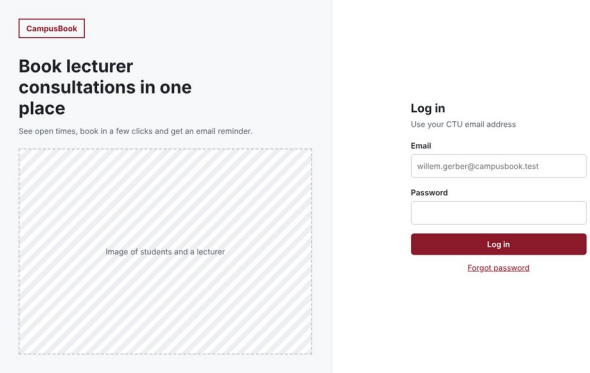


Figure 4.17 *Desktop login*

Wrong login details

The message shown when the email or password is wrong

The user can try again

The message does not say which one was wrong

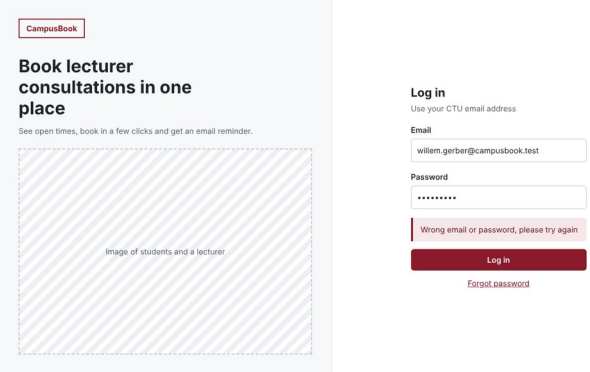


Figure 4.18 *Wrong login details*

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Session expired

When a session ends the user goes back to the login page with a message to log in again

This makes sure protected pages need a valid login

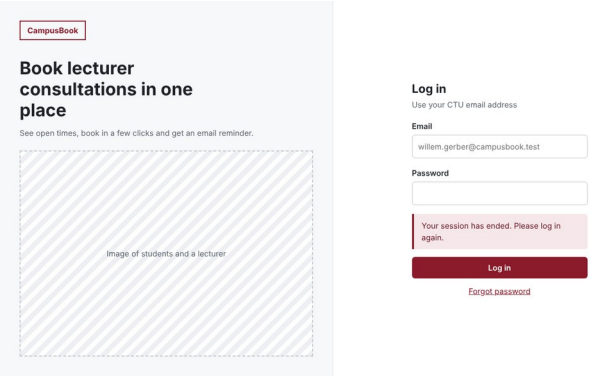


Figure 4.19 *Session expired message*

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Mobile pages

The login booking calendar and booking confirmation pages on a phone

The layout stacks vertically so the form and the times stay easy to use on a small screen

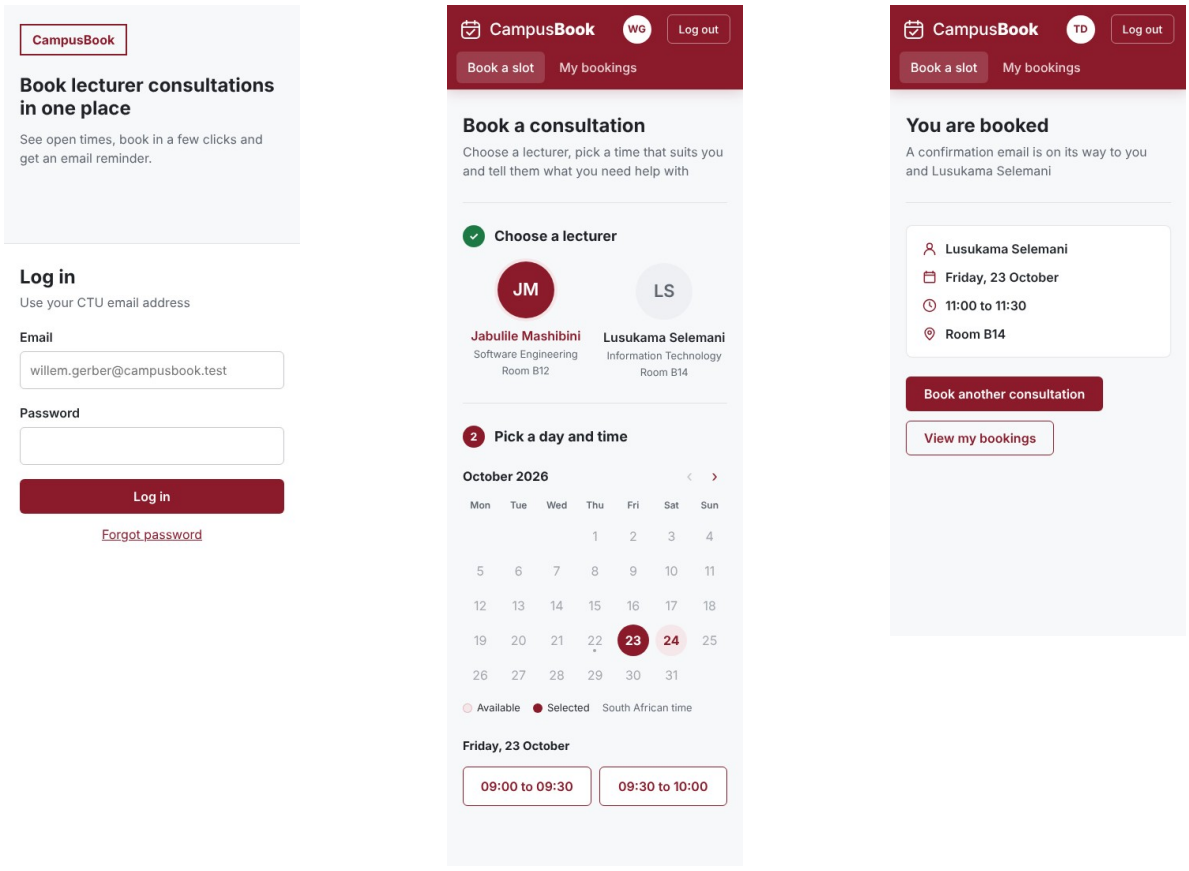


Figure 4.20 *Mobile login, booking calendar and confirmation pages*

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Student booking

Desktop booking page

The student picks a lecturer a date and a time

The summary shows the lecturer date time and location and a reason box explains the purpose before the booking is confirmed

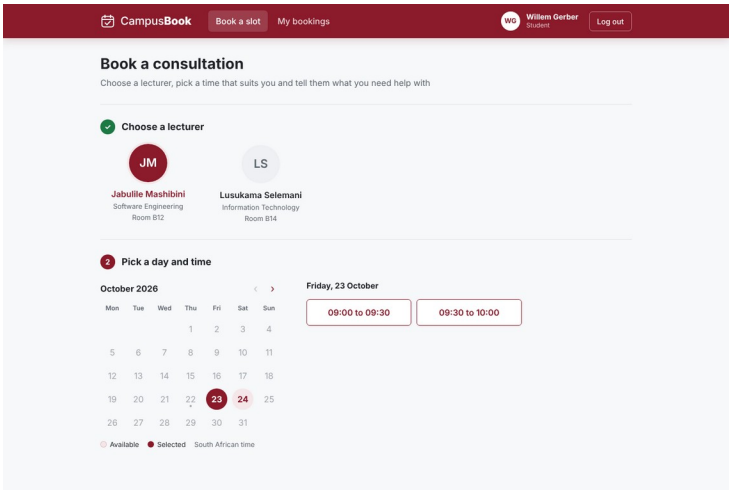


Figure 4.21 *Desktop booking page*

Times displayed correctly

The available times show correctly and the booking summary matches the selected slot

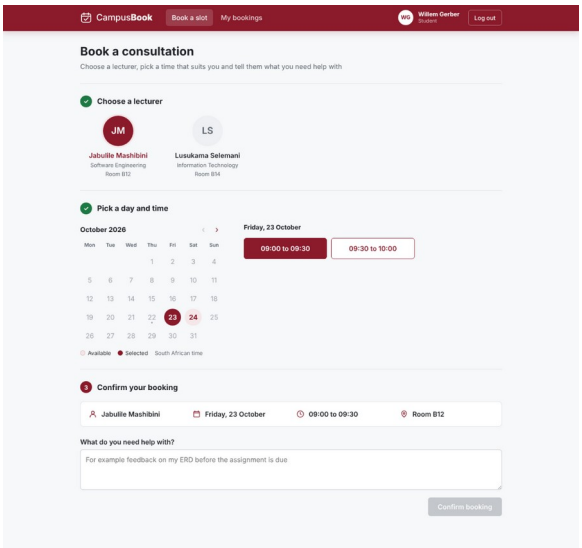


Figure 4.22 *Times displayed correctly*

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Availability after a booking

After a slot is taken only the free time is still offered for that date

The form keeps the reason the student typed

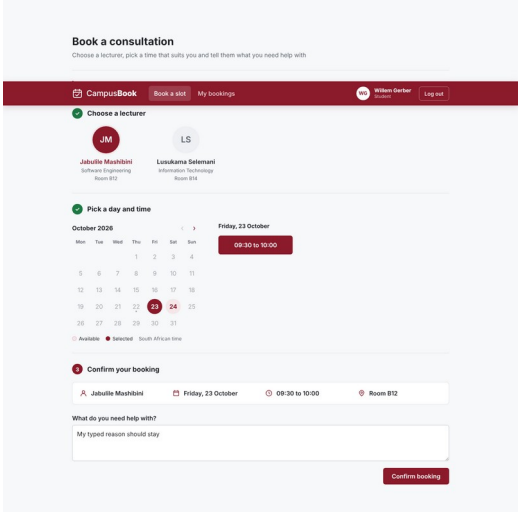


Figure 4.23 *Availability after a booking*

My bookings

Desktop my bookings

Upcoming bookings and history are shown in two sections with a button to book again

When there are none a message says so

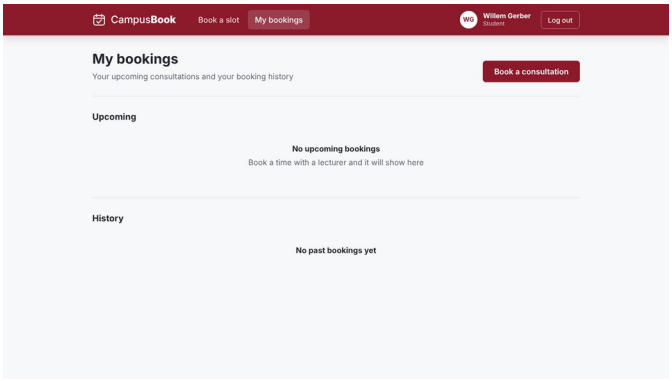


Figure 4.24 *Desktop My bookings*

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Mobile my bookings

The same page on a phone with upcoming and past sections and a button for a new booking

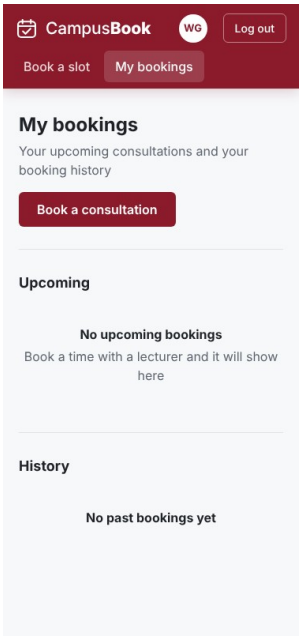


Figure 4.25 *My bookings on a phone*

Moving a booking

The student picks another time from the dropdown and presses Move Booking to change an existing consultation

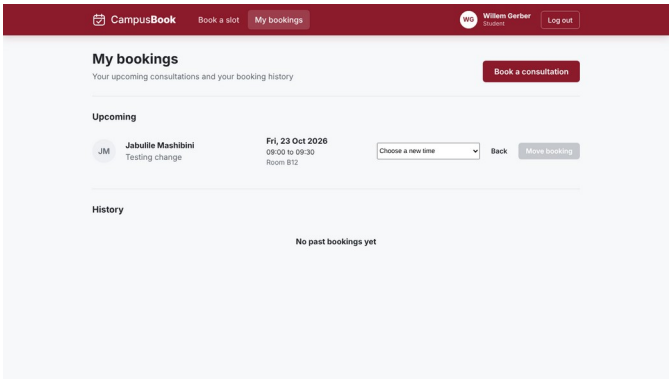


Figure 4.26 *Moving a booking*

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Lecturer slots

Slot published

The lecturer page after slots are published

The table shows date time location and status with Edit and Remove buttons

A green message confirms the new slot

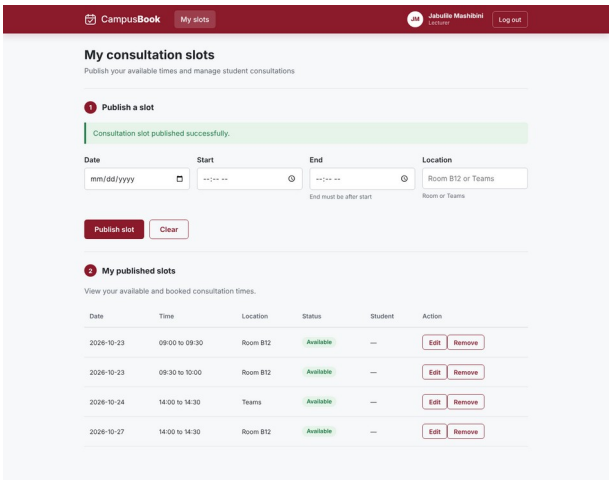


Figure 4.27 *Slot published*

Slot updated

A slot for 27 October 2026 was changed to Room C3

A green message confirms the update and the table shows the new location

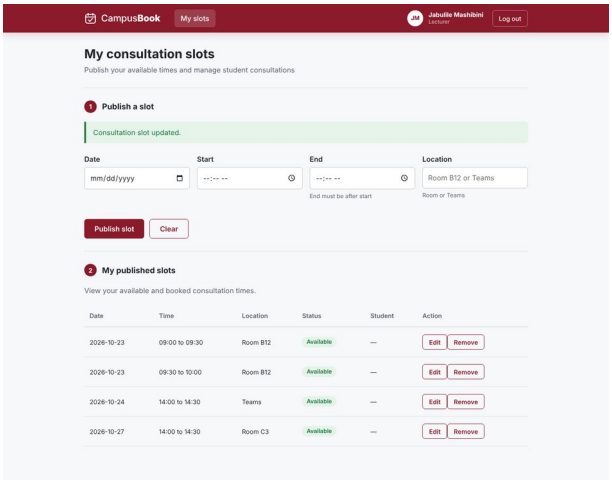


Figure 4.28 *Slot updated*

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Errors found and fixed

Time display error

Earlier the time buttons only showed the word to and the summary showed undefined to undefined

The time values were missing in the frontend

This was found in testing and fixed

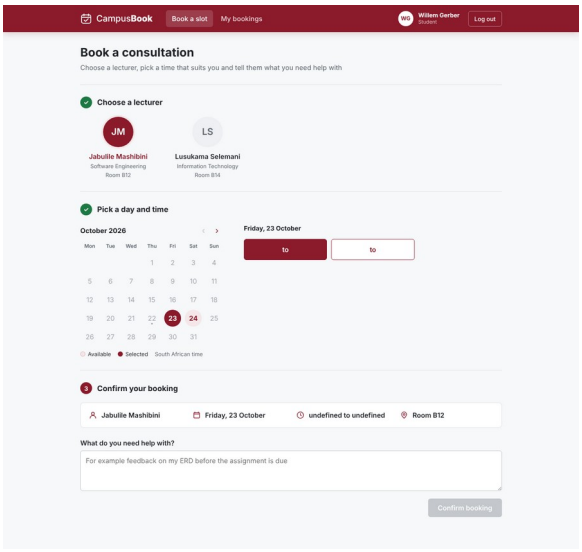


Figure 4.29 *Time display error*

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Slot route error

The page showed that the slot route did not exist

The lecturer page still opened but slots could not be loaded or published

This was a backend integration issue and it was corrected

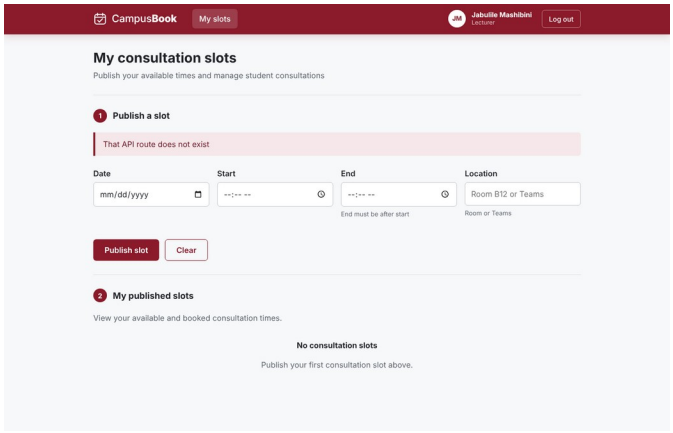
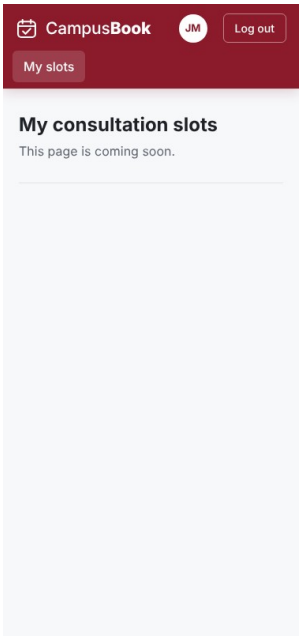


Figure 4.30 *Slot route error*

Earlier lecturer page

Lecturer page on mobile before slots

An early phone view with the navigation and a coming soon message before slot management was built



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Figure 4.31 *Lecturer page on a phone before the slots were added*

Lecturer page on desktop before slots

An early desktop view with the same coming soon message

It shows the layout before the slot features were added

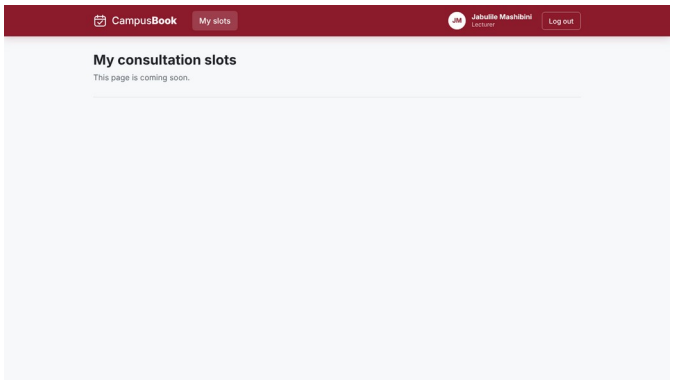


Figure 4.32 *Lecturer page on a desktop before the slots were added*

Summary of 4.7

The screenshots show login booking changing bookings and slot management working together between students and lecturers

They also show errors that were found and fixed during development

4.8 Version Control and Team Development

GitHub was used to manage development of the CampusBook lecturer consultation booking system. The team worked in a shared repository and used branches and pull requests to organise changes, review contributions and merge work into shared branches. Commit history and pull-request records provide evidence of how the application developed over time and how team members contributed.

GitHub repository overview

The repository overview provides a central location for the project source code, branches, commits and pull requests.

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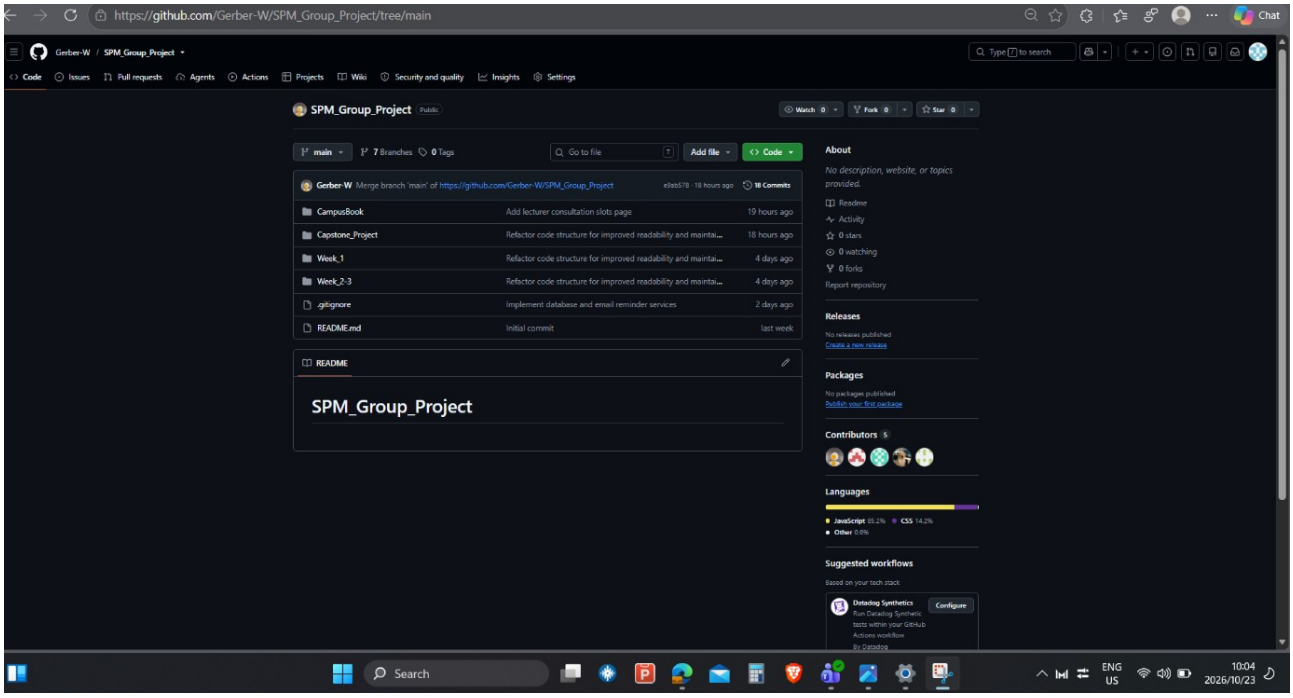


Figure 4.33 *GitHub repository overview*

Pull Request #4: Slots API Routes

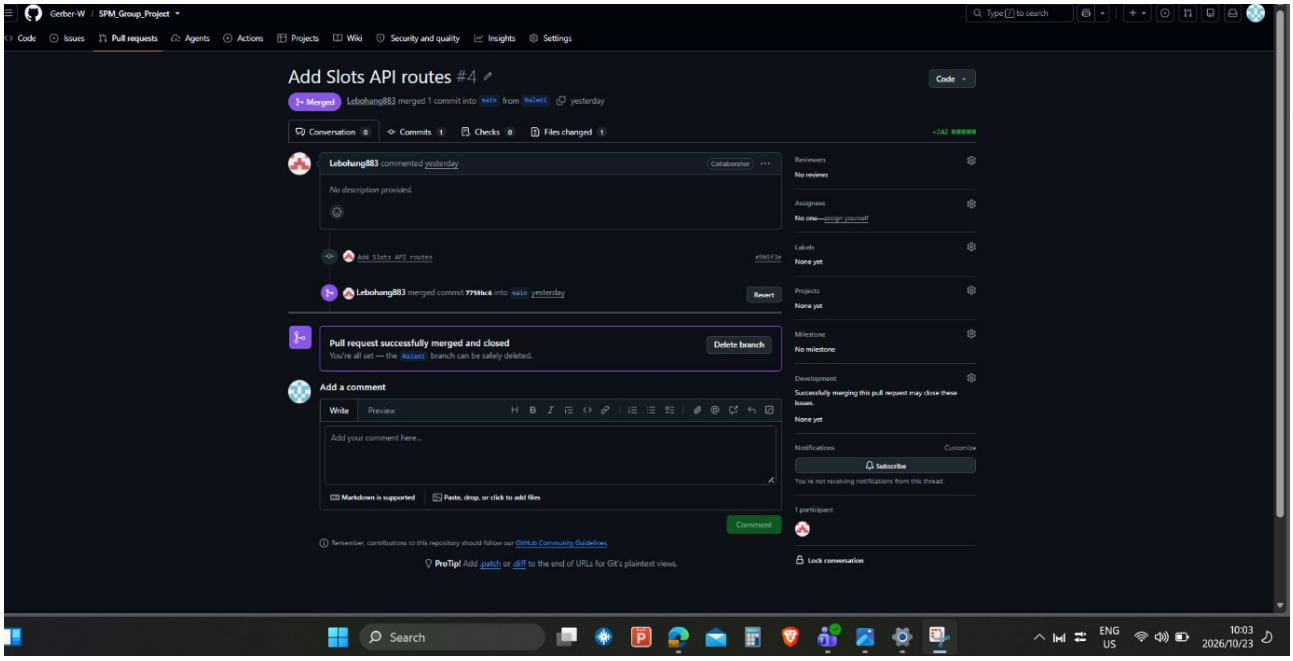


Figure 4.34 *Pull Request #4 Slots API routes*

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Pull Request #4, titled “Add Slots API routes,” submitted changes from the Naledi branch and was merged into main by Lebohang883. The pull request records one commit and one changed file. It provides evidence of the Slots API being added to the shared codebase.

Pull Request #6: Lecturer Consultation Slots

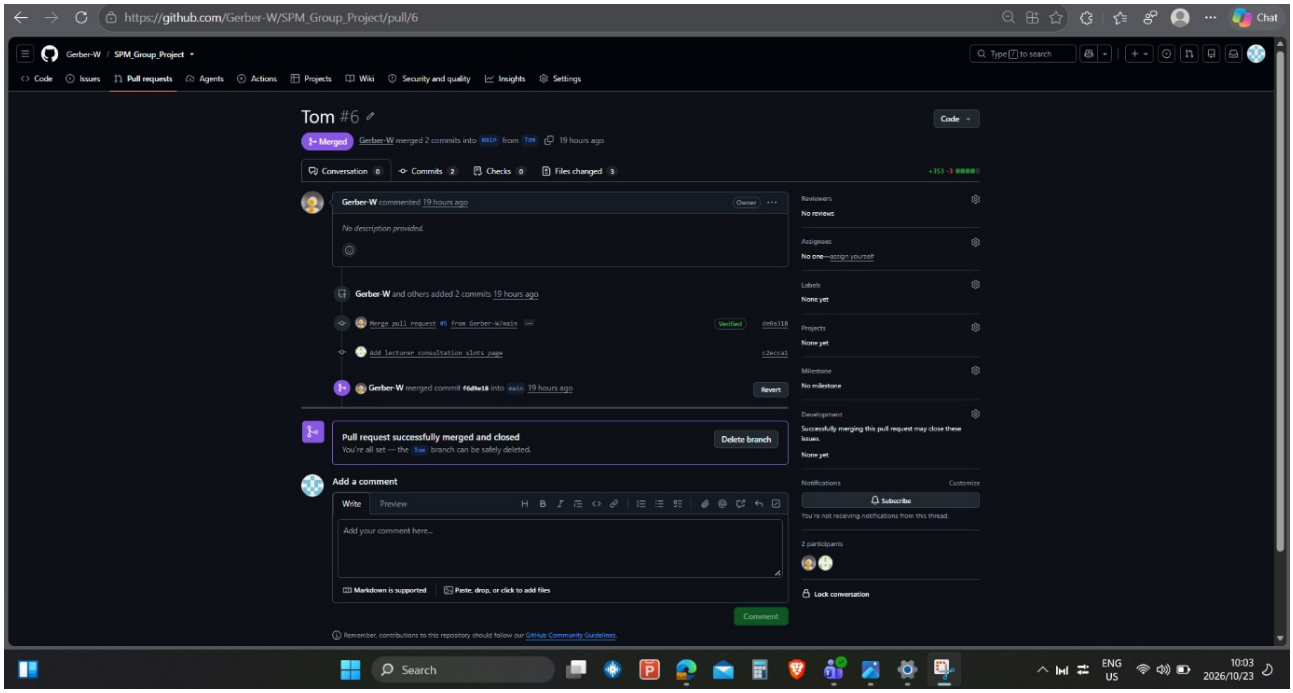


Figure 4.35 *Pull Request #6 Lecturer consultation slots*

Pull Request #6, titled “Tom,” records two commits from the Tom branch merged into main by Gerber-W. The pull request shows three changed files and provides evidence of development work on the lecturer consultation-slots page.

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Pull Request #3: Database and Email Reminder Services

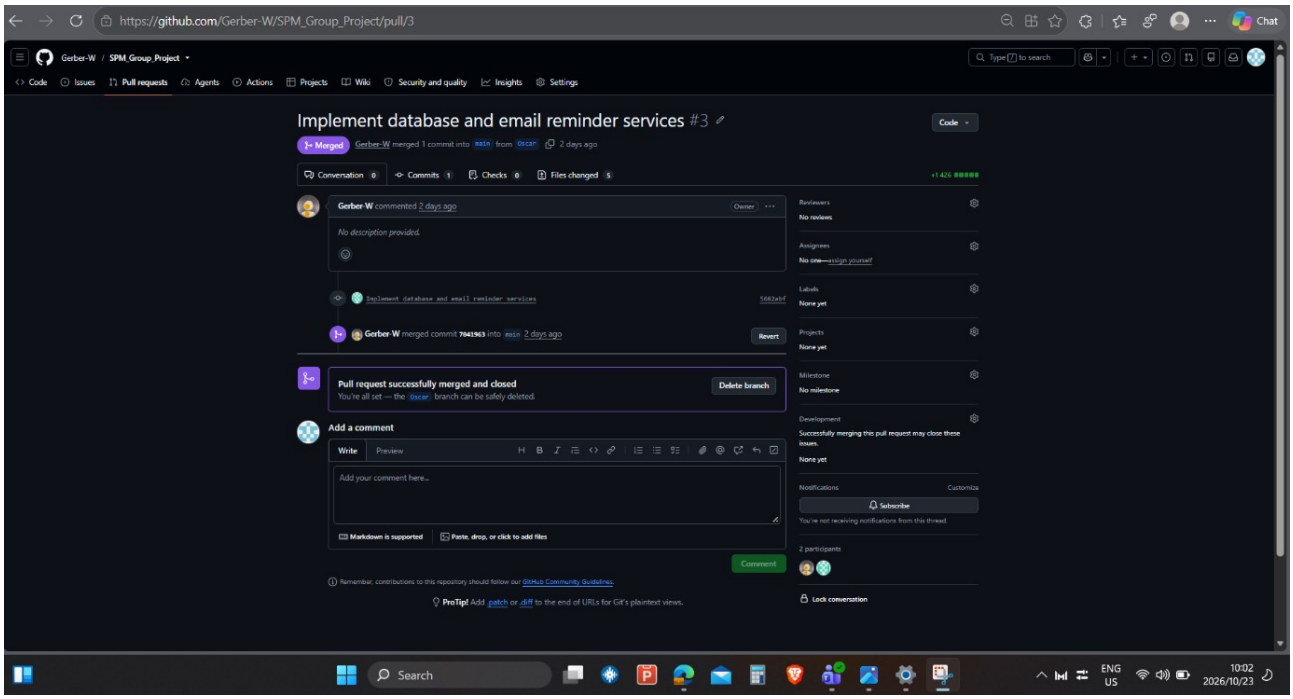


Figure 4.36 *Pull Request #3 Database and email reminder services*

Database and email work is documented in Pull Request #3, which includes commit 5682abf, “Implement database and email reminder services,” from the Oscar branch. The work includes the reusable MySQL connection module (db.js), the mailer service (mailer.js) and the reminder scheduler (reminders.js). This provides traceable evidence of the implementation through both a pull request and a descriptive commit.

Branches Overview

The repository's branches, commit history and pull requests provide evidence of the team's development activity and individual contributions. The records also show that the team used branches and pull requests to integrate changes into the main branch.

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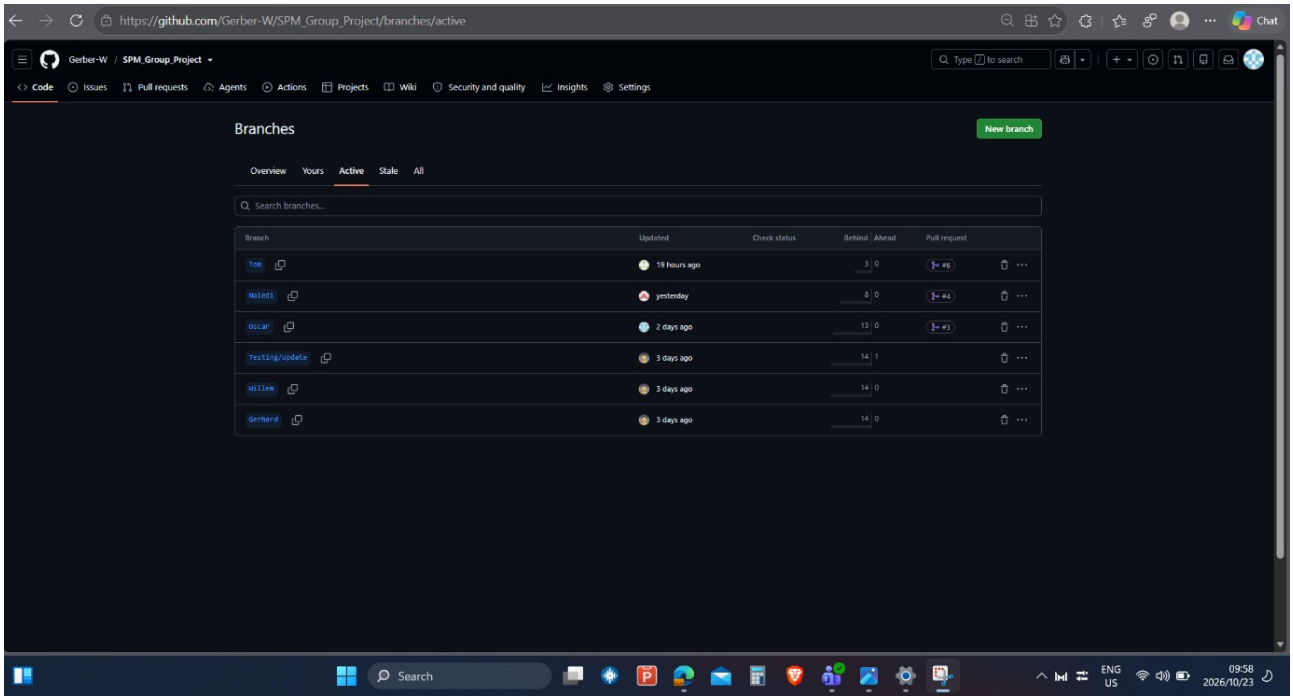
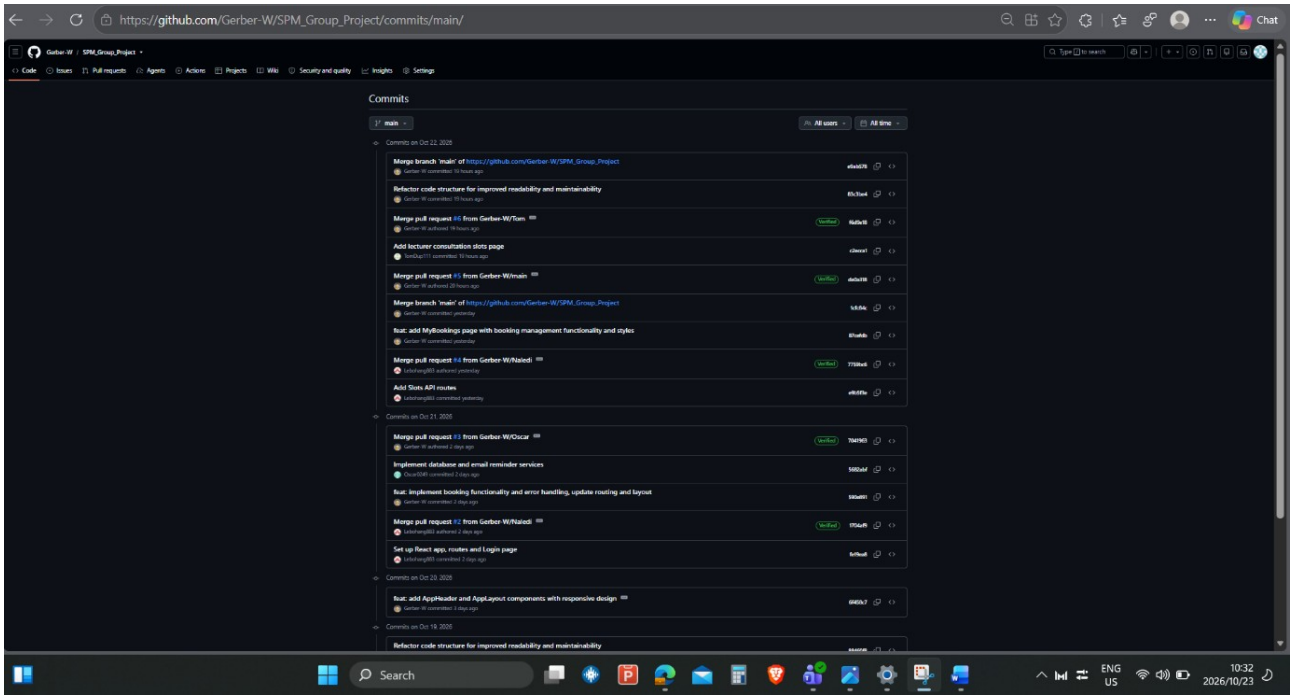


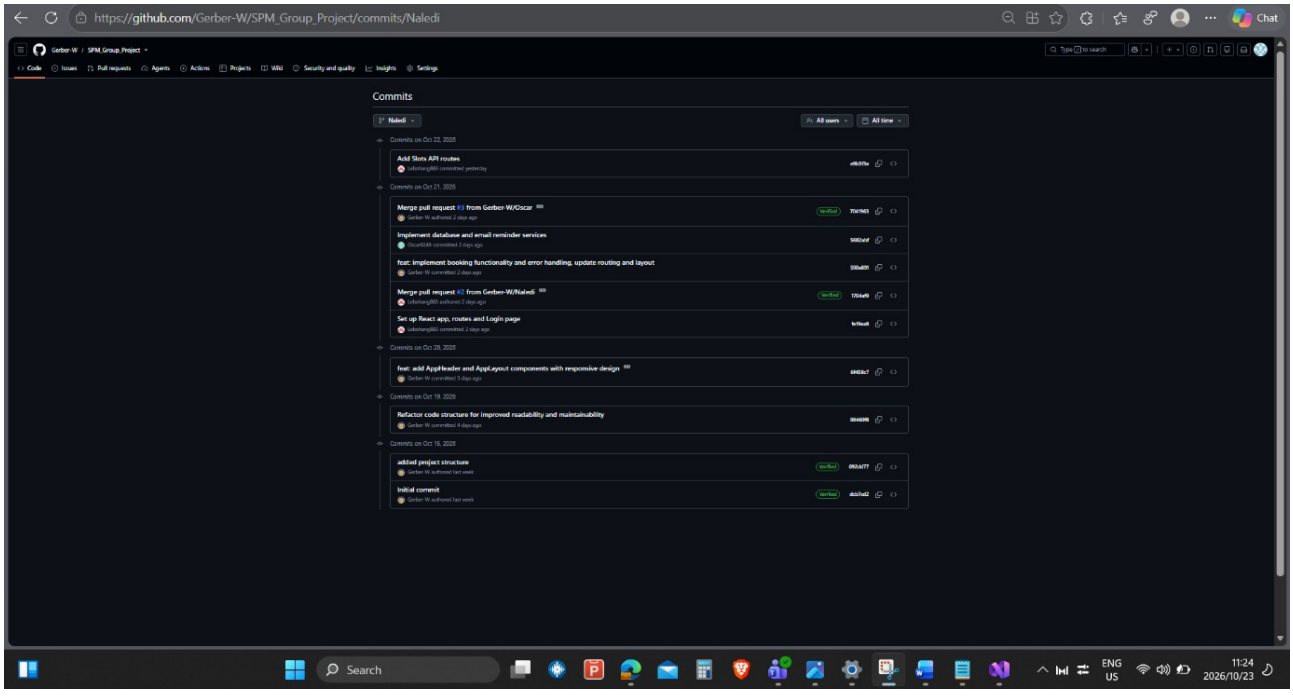
Figure 4.37 *Branches overview*

Overall Commit History

These commits recorded changes to the CampusBook project through descriptive commit messages and timestamps. The repository also contained seven branches, allowing team members to organise their development work separately. The commit history provided evidence of project progress and helped identify individual contributions and the integration of changes into the main branch.

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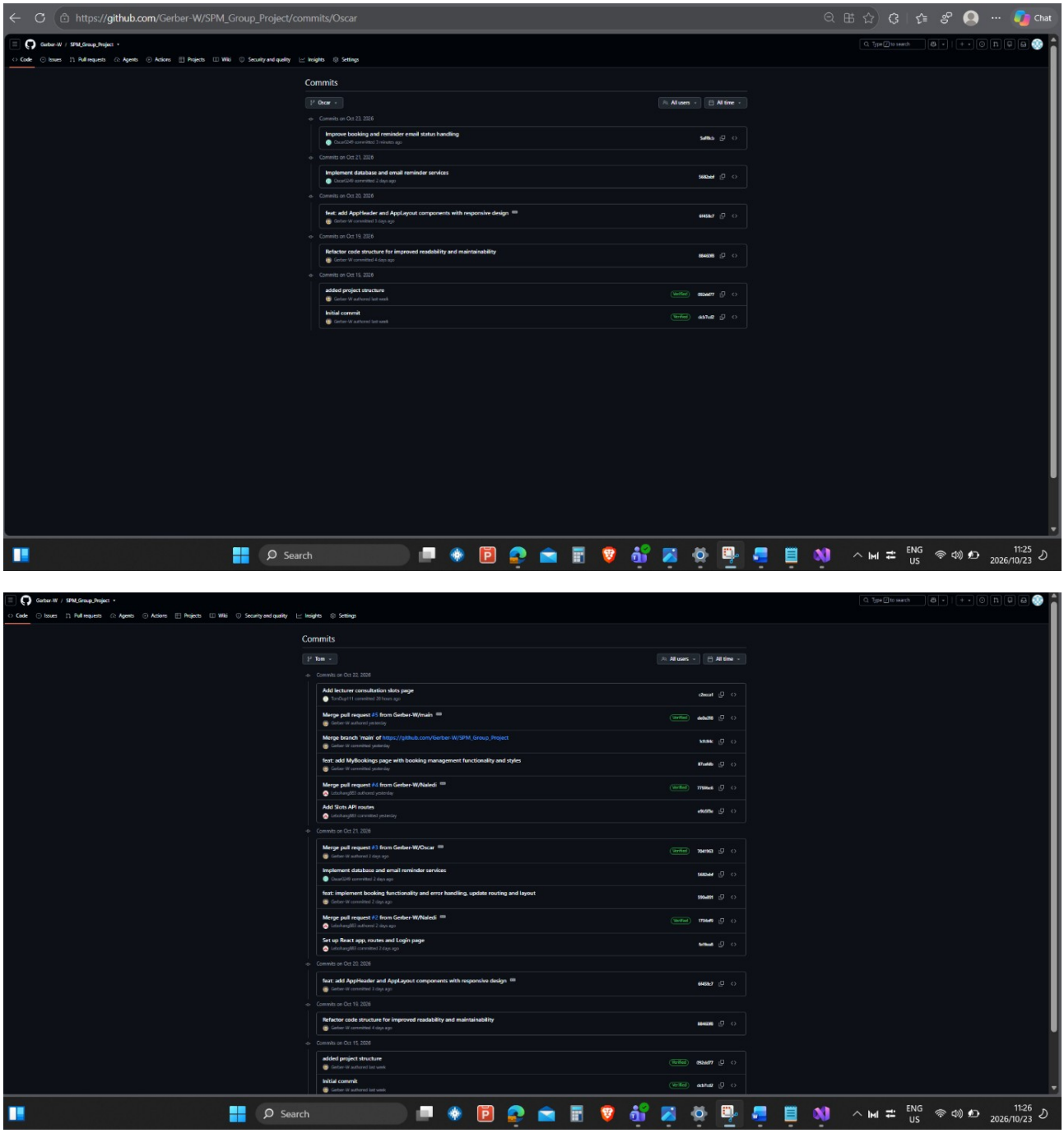


Figure 4.38 *Overall commit history*

Commits and Pull Requests

The source code of CampusBook is kept in one GitHub repository with a main branch that always holds the working version and a branch for each member

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Members commit their own work and open a pull request and another member reviews it before it is merged into main

This keeps the history clear and shows who did what

Table 4.4 *Key commits and pull requests*

Commit or pull request	Date	Member	What it holds
dcb7cd2 and 092dd77	15 Oct	Willem Gerber	Repository and project structure
88460f8	19 Oct	Willem Gerber	Back-end folders and schema and seed scripts and diagrams
6f458c7 and pull request #1	20 Oct	Willem Gerber	Back-end set up and login and middleware and Bookings API
fe19ea8 and pull request #2	21 Oct	Lebohang Tlaitlai	React set up and Login page
5682abf and pull request #3	21 Oct	Oscar Khumalo	Database connection and email reminders
590e891	21 Oct	Willem Gerber	Booking error handling and Book a slot page
e9b5f3e and pull request #4	22 Oct	Lebohang Tlaitlai	Slots API
87cefdb	22 Oct	Willem Gerber	My bookings page and phone layout
c2ecca1 and pull request #6	22 Oct	Thomas du Plessis	Lecturer slots page

The fixes found in testing were committed through a pull request and merged into main after review

Chapter Summary

Chapter 4 showed how CampusBook was designed and built

The architecture and the diagrams and the wireframes were turned into a React front end and an Express back end with a MySQL database

All the planned pages work for students and lecturers and the screenshots in 4.7 show them running

Version control with GitHub kept the work of the group clear

Chapter 5 tests the system against the requirements and reports the results

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Chapter 5 Testing, Results and Evaluation

5.1 Test Plan

This chapter shows how CampusBook was tested and what the tests found It checks the working app against the functional and non-functional requirements from Chapter 3

The tests were run on 22 October 2026 on the main branch at commit e9ab578 It contains the login Slots API Bookings API database and email services the Book a slot page the My bookings page and the Lecturer slots page that Thomas merged on 22 October The tests found defects so the fixes were made and every failed test was run again The results show the retest and the failure before the fix

This chapter includes the student survey results and the user testing with 4 students and 2 lecturers on 28 October 2026

Test Strategy

Testing followed the Test Plan in folder 06 Tests were written for every requirement before they were run and each test has an expected result and an actual result in the Test Cases and Results workbook

Table 5.1 *Test levels*

Level	What was tested	Tool
API tests	Every route with valid and invalid input and wrong roles	Node script
Integration tests	API and database and the reminder job and the email preview	Node script and SQL
Browser tests	Pages and journeys and route protection at 375 and 1366 pixels	Playwright and Chromium
Security checks	Role checks and forged tokens and other users records and password storage	Node script and SQL
Performance checks	Page load and API response times	Browser and script timings

Table 5.2 *Test environment*

Item	Detail
Application	Branch main at commit e9ab578
Back end and database	Node.js 22 and Express with MariaDB 10.11 built from schema.sql and seed.sql
Front end	React 18 and Vite 5 production build
Browser	Chromium through Playwright at 375 x 800 and 1366 x 768
Test data	Seven made up users on the campusbook.test domain
Email	Preview mode so emails print to the server log

The MariaDB server was used in place of MySQL because it runs the same SQL and the same transactions and row locks A final run on MySQL on the demonstration laptop gave the same results

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The browser tests were run on the production build Every failed test was first run on the merged code to record the defect and then run again after the fix The fixes were then committed and pushed to the main branch on GitHub

Willem Gerber led the testing and ran the tests Each developer checks and retests the defects in their own files

Acceptance criteria

- Every Must requirement has a test with an actual result
- Every failed test has a defect with an owner
- No open defect stops a student from booking a consultation

5.2 Test Cases and Results

36 test cases were written and linked to the requirements in the Requirements Traceability Matrix The full steps test data and actual results are in Group-3_Test_Cases_and_Results.xlsx

Table 5.3 *Summary of test results*

Result after the fixes	Number of tests
Pass	35
Fail	0
Not run (bookings not timed in the user testing)	1
Total	36

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Table 5.4 *All test cases and results*

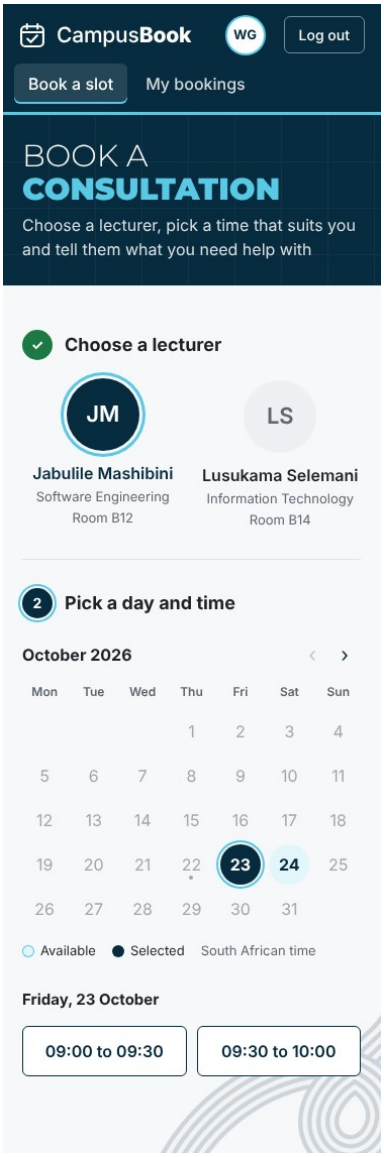
Test	Requirement	What was tested	Result
TC-01	FR-01	Login	Pass
TC-02	FR-02	Role checks on the server	Pass
TC-03	FR-03	Lecturer manages own slots	Pass
TC-04	FR-04	Student views open slots	Pass
TC-05	FR-05	Book a slot	Pass
TC-06	FR-06	Change and cancel a booking	Pass
TC-07	FR-07	No double booking	Pass
TC-08a	FR-08	Confirmation email	Pass
TC-08b	FR-08	Reminder email	Pass
TC-09	FR-09	Booking history	Pass
TC-10	NFR-01	First time student books in under 2 minutes	Not Run
TC-11	NFR-02	Main pages load in under 3 seconds	Pass
TC-12	NFR-03	Phone and laptop layout	Pass
TC-13	NFR-04	Passwords stored as bcrypt hashes	Pass
TC-14	NFR-05	Users cannot reach other records	Pass
TC-15	NFR-06	Only needed data is stored	Pass
TC-16	NFR-07	Errors explain what to fix (API)	Pass
TC-16b	NFR-07	Errors on screen and kept input	Pass
TC-17	FR-05	Book a missing past or taken slot	Pass
TC-18	FR-03	Slot validation rules	Pass
TC-19	FR-03	Booked slot cannot be edited or removed	Pass
TC-20	FR-03	Removing a slot keeps history	Pass
TC-21	FR-02	Bad ids bad JSON and unknown routes	Pass
TC-22	FR-01	Forged expired and garbage tokens	Pass
TC-23	NFR-02	Health check	Pass
TC-24	NFR-02	API response time for the slot list	Pass
TC-25	FR-03 and FR-09	Lecturer publishes a slot from the page	Pass
TC-26	FR-08	Reminder not sent for a cancelled booking	Pass
TC-27	FR-02	Pages protected by login and role	Pass
TC-28	NFR-07	Expired or invalid session	Pass
TC-29	FR-06	Change a booking from the page	Pass
TC-30	Security (extra)	Repeated wrong passwords are limited	Pass
TC-31	FR-07	Database refuses a second confirmed booking	Pass
TC-32	FR-04	Times shown on the Book a slot page	Pass
TC-33	FR-03	Lecturer removes a slot from the page	Pass
TC-34	FR-03	Lecturer edits a slot from the page	Pass

Key results

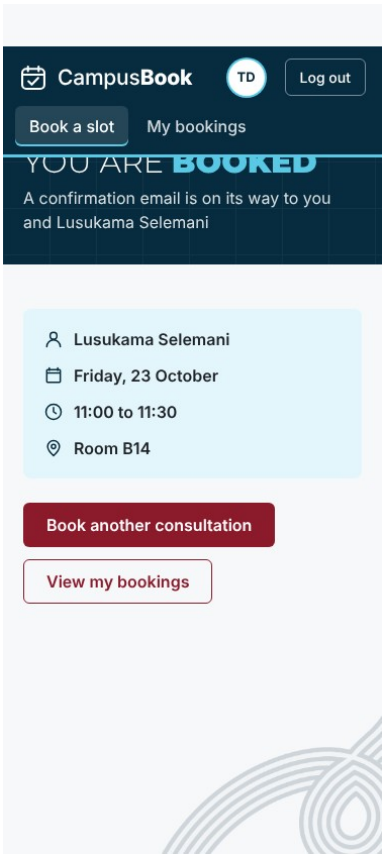
- Double booking (TC-07) Eight booking requests for one slot were sent at the same moment from three students One booking was created and the other seven were refused with the message Sorry that slot is no longer available The database also refuses a second confirmed booking on its own (TC-31)
- Access control (TC-02 and TC-14) Wrong roles get 403 and missing tokens get 401 and a student cannot cancel or change another student’s booking
- Emails (TC-08a and TC-08b) The confirmation goes to the student and the lecturer and a due reminder is sent and marked sent A cancelled booking leaves no reminder to send (TC-26)
- Passwords (TC-13) All seven stored passwords are 60 character bcrypt hashes

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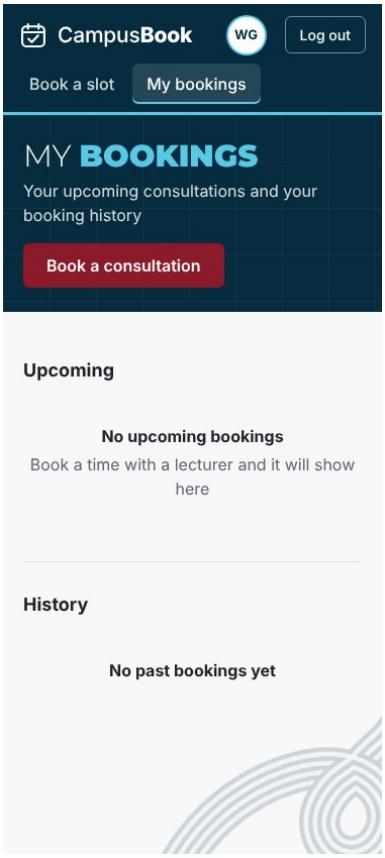
- Slot rules (TC-18 to TC-20) Bad times and past dates and overlaps are refused and a booked slot cannot be edited or removed and removing a slot keeps its booking history



Book a slot on a phone



Booking confirmed on a phone



My bookings on a phone

Figure 5.1 *Booking a slot on a phone*

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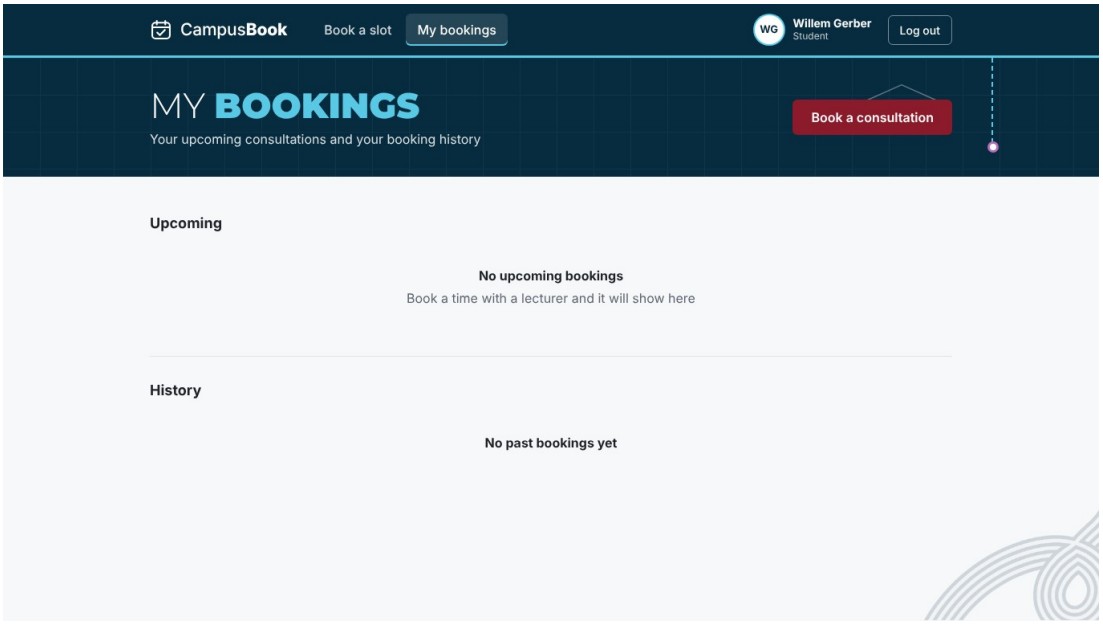


Figure 5.2 *My bookings page with no bookings*

5.3 Defect Log

On the merged code 6 tests failed and one was blocked and every failure is logged in Group-3_Defect_Log.xlsx 10 defects were logged in total 1 was fixed earlier 9 were fixed after testing and passed the retest and 0 are open Tables 5.3 and 5.4 show the results after the fixes

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Table 5.5 *Defect summary*

ID	Feature	Defect	Severity	Status
D-01	Bookings API and My bookings	Booking dates came back as time zone shifted dates because the raw slot_date was returned so a booking could show on the wrong day	High	Closed
D-02	Book a slot page	Time buttons show only "to" and the confirm panel shows "undefined to undefined" because the page reads start and end but the Slots API sends startTime and endTime	High	Fixed
D-03	Lecturer slots page	The merged page does not work with the Slots API: it calls /api/slots/my but the route is /api/slots/mine so the list shows That API route does not exist and it sends start and end but the API needs startTime and endTime so publishing fails (FR-03 and FR-09)	High	Fixed
D-04	My bookings page	No change button although the API supports changing a booking (FR-06)	Medium	Fixed
D-05	Session handling	An expired or invalid token leaves the user on the page with an error and the old token is kept instead of going to the login page	Medium	Fixed
D-06	Login security	There is no limit on repeated wrong passwords so 40 attempts in a row were all answered	Medium	Fixed
D-07	Documentation	Chapter 4 says the reminder scheduler runs every 15 minutes but the code runs it every minute	Low	Fixed
D-08	Book a slot page	After a failed booking the reason box is hidden until a time is chosen again although the typed text is kept	Low	Fixed
D-10	Lecturer slots page	The page can publish and remove slots but has no way to edit a slot although the API supports editing (FR-03)	Medium	Fixed
D-09	Route protection	A logged in user who opens the other roles page is sent to the login page and not to their own home page	Low	Fixed

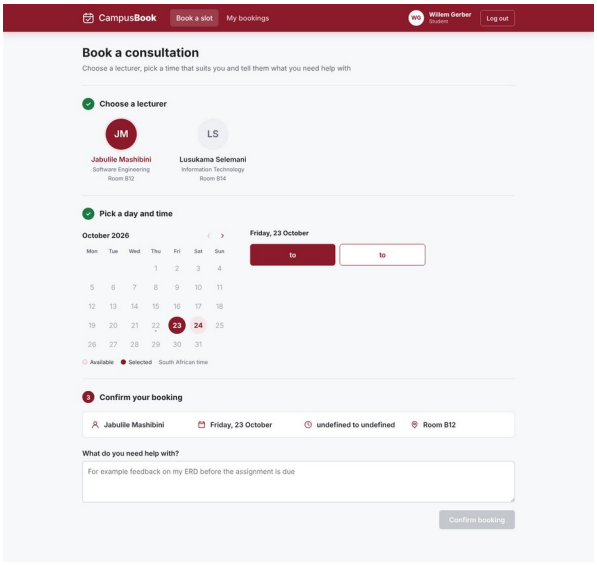
D-01 Booking dates shifted by the time zone (closed)

While building My bookings the dates of some bookings were one day out The query returned the raw database date and the server turned it into a date with a time zone so the day could move The query now returns the date as text with DATE_FORMAT (commit 87cefdb) The retest shows the same date that is stored in the database (TC-05 and TC-09)

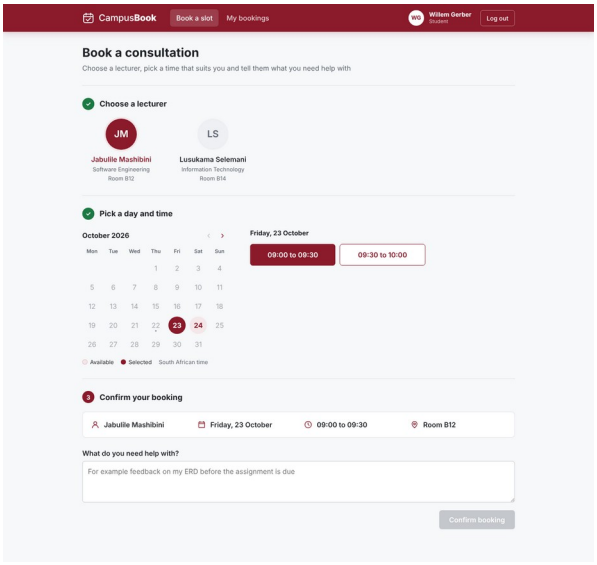
D-02 Missing times on the Book a slot page (fixed)

The browser test showed time buttons that only said to and a confirm panel that said undefined to undefined The Book a slot page reads start and end but the Slots API sends startTime and endTime so the page had nothing to show The booking itself still worked The fix maps the two names when the slots are loaded and is three lines in BookSlot.jsx With the fix the buttons show 09:00 to 09:30 and 09:30 to 10:00 The retest passed

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Before the fix



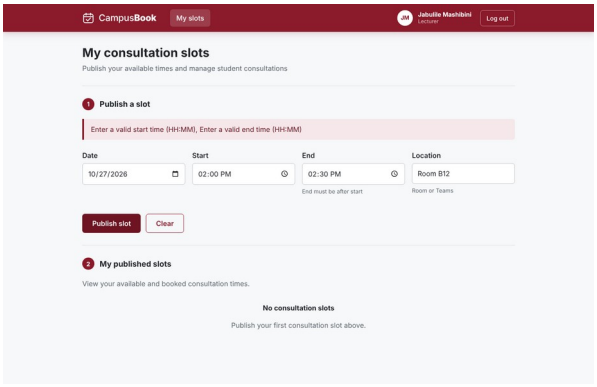
After the fix

Figure 5.3 *Time buttons before and after the D-02 fix*

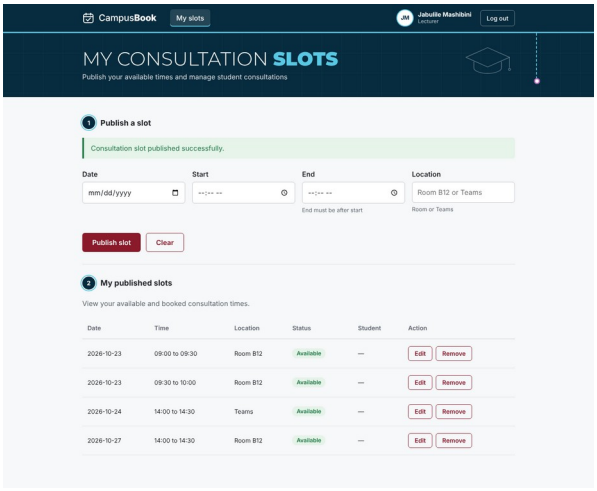
D-03 Lecturer slots page does not work with the Slots API (fixed)

Thomas merged the Lecturer slots page on 22 October When it was tested the page showed That API route does not exist because it asks for /api/slots/my and the route is /api/slots/mine Publishing a slot failed with Enter a valid start time because the page sends start and end and the API needs startTime and endTime Nothing was saved The fix is four lines in LecturerSlots.jsx With the fix the page lists the slots and publishing and removing a slot both work The retest passed and an Edit button was also added for D-10

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Merged code



With the fix

Figure 5.4 *Lecturer slots page before and after the D-03 fix*

D-04 to D-10 other defects (fixed)

- D-04 My bookings had no change button so a Change time button was added that lists the other open times of that lecturer and moves the booking (TC-29)
- D-05 An expired session stayed on the page so the app now clears the login and shows the login page with the message Your session has ended (TC-28)
- D-06 Repeated wrong passwords were not limited so the login now answers 429 after 10 wrong passwords in 15 minutes and other accounts are not affected (TC-30)
- D-08 The reason box was hidden after a failed booking so the confirm step now stays on screen unless the slot was taken (TC-16b)
- D-09 A user opening the other roles page was sent to the login page so they now go to their own home page (TC-27)
- D-10 The Lecturer slots page had no edit option so an Edit button was added (TC-34)
- D-07 The Chapter 4 text said the reminder job runs every 15 minutes but the code runs it every minute so the text now says every minute and the label in the architecture diagram (Figure 4.1) now matches the code

Open defects

- No defects are open and the fixed code is on the main branch on GitHub

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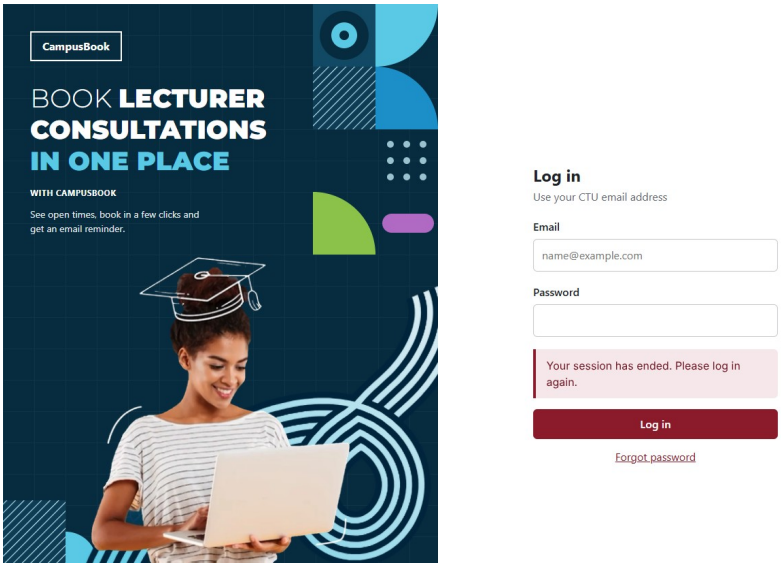


Figure 5.5 *Session ended message after the D-05 fix*

5.4 Non-Functional Requirement Evaluation

Table 5.6 *Performance and layout measurements*

Check	Target	Measured	Result
Login page load (phone and laptop)	Under 3000 ms	78 ms and 78 ms	Pass
Book page after login	Under 3000 ms	248 ms and 220 ms	Pass
My bookings page	Under 3000 ms	380 ms and 372 ms	Pass
API slot list over 20 requests	Under 500 ms	Median 5 ms and slowest 8 ms	Pass
Sideways scrolling at 375 and 1366 pixels	None on every page	None on all 5 pages tested	Pass
Scripted booking journey on a phone	Under 120 seconds	1.8 seconds	Indicative only

The times were measured on the team test set up with the API and the database on the same machine so they leave out the network delay of a real campus connection They show that the pages themselves are light The built page script is about 239 kB

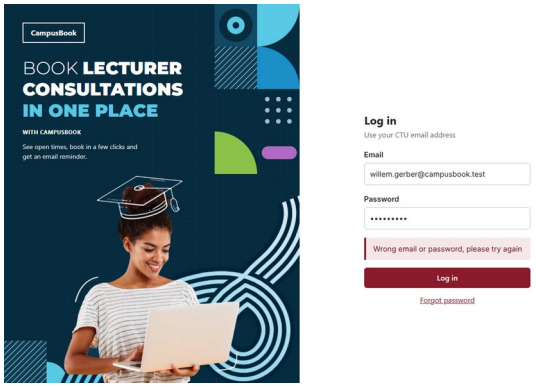
The scripted journey shows that the booking flow has few steps and no waiting The real answer for NFR-01 needs first time students and a stopwatch so the timed test was planned for the user testing but the bookings were not timed there (section 5.6)

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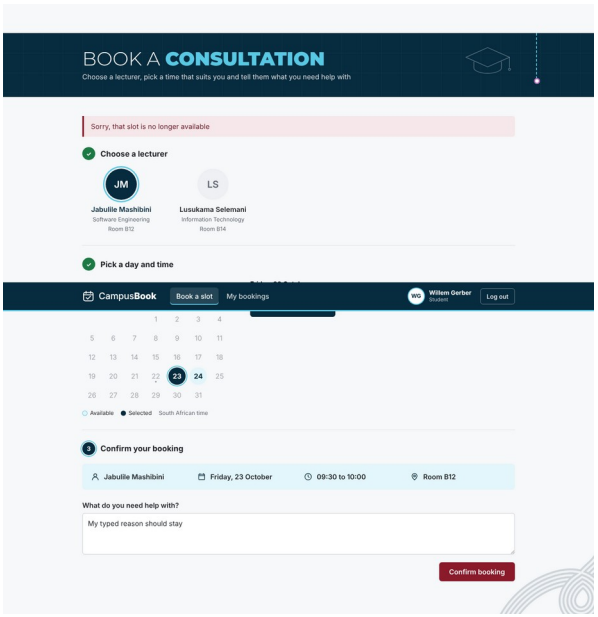
Usability check against the ten heuristics (Nielsen, 1994)

Table 5.7 *Heuristic check (Nielsen, 1994)*

Heuristic	Evidence from the app
Visibility of system status	A confirmation page after booking and a success notice after cancelling and the booking steps are numbered
Match with the real world	Plain words such as Book a consultation and Cancel booking and dates written out in full
User control and freedom	Cancelling asks for a second click with Keep it and Yes cancel
Error prevention	The confirm button stays disabled until a reason is typed and booked slots cannot be picked
Help users recover from errors	Messages say what to fix such as Enter a valid email address and Sorry that slot is no longer available
Consistency	One theme and the same buttons and alerts on every page
Problems found and fixed	No change button (D-04) and a hidden reason box after an error (D-08) and no redirect on an expired session (D-05)



Login error with the email kept



Slot taken message

Figure 5.6 *Login error and slot taken messages*

5.5 Requirements Traceability Matrix

The traceability matrix links every requirement from Chapter 3 to the part of the system that delivers it and the test cases that check it so no requirement is left without a test The full matrix is in Group-3_Requirements_Traceability_Matrix.xlsx in folder 06

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Table 5.8 *Requirements traceability matrix*

ID	Requirement	Design or component	Test cases	Result
FR-01	Students and lecturers log in with their email and password	routes/auth.js and Login.jsx	TC-01 and TC-22	Pass
FR-02	The system checks the user role so students and lecturers only reach their own pages and actions	middleware/auth.js and ProtectedRoute.jsx	TC-02 and TC-21 and TC-27	Pass
FR-03	Lecturers publish edit and remove their own future consultation slots	routes/slots.js and LecturerSlots.jsx	TC-03 and TC-18 and TC-19 and TC-20 and TC-25 and TC-33 and TC-34	Pass
FR-04	Students view the open slots of a chosen lecturer	routes/slots.js and BookSlot.jsx	TC-04 and TC-32	Pass
FR-05	Students book an open slot and give a short reason for the visit	routes/bookings.js and BookSlot.jsx	TC-05 and TC-17	Pass
FR-06	Students change or cancel their own upcoming bookings	routes/bookings.js and MyBookings.jsx	TC-06 and TC-29	Pass
FR-07	The system stops two students booking the same slot	bookings.js transaction and unique active_slot_id	TC-07 and TC-31	Pass
FR-08	The system sends an email confirmation after booking and a reminder before the meeting	services/mailer.js and jobs/reminders.js	TC-08a and TC-08b and TC-26	Pass
FR-09	Students see their own booking history and lecturers see the bookings for their slots	routes/bookings.js and MyBookings.jsx	TC-09 and TC-25	Pass
NFR-01	A first-time student completes a booking in under 2 minutes	BookSlot.jsx three step flow	TC-10	Not run
NFR-02	The main pages load within 3 seconds in the team test set up	Vite production build	TC-11 and TC-23 and TC-24	Pass
NFR-03	Every page works on a phone at 375 pixels wide and on a laptop at 1366 pixels wide	CSS container queries and responsive header	TC-12	Pass
NFR-04	Passwords are stored as bcrypt hashes and never as plain text	routes/auth.js and the seed script	TC-13 and TC-30	Pass
NFR-05	Each user can only see their own bookings or the bookings for their own slots	user_id checks in every query	TC-14 and TC-09	Pass
NFR-06	Only the data needed for logging in and booking is stored with made-up data during development	database/schema.sql and seed.sql	TC-15	Pass
NFR-07	Errors explain what to fix and the form keeps what the user already typed	middleware/validate.js and the page alerts	TC-16 and TC-16b and TC-28	Pass

Every functional and non-functional requirement is traced to at least one test case and each test case is traced back to a requirement in Table 5.4

5.6 Stakeholder and User Feedback

The student survey was answered by 13 students on 22 October 2026 The survey was anonymous and the raw answers are in Group-3_Survey_Responses.xlsx in folder 06 The target of at least 10 students was met

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Table 5.9 *Student survey results (13 students)*

Question	Result
Needed to meet a lecturer in the last 3 months	11 of 13 needed to at least once (10 said 1 to 2 times and 1 said 3 to 5 times)
How they arrange a consultation now	Teams 11 and asking in class 5 and going to the office 3 and WhatsApp 2
How easy it is now (1 hard to 5 easy)	Average 3.2 and 4 of 13 gave 1 or 2
Gave up on a consultation because of the problems	7 of 13 said yes
Time from first asking to a confirmed time	1 to 2 days for 10 of 13 and the same day for 3
Hardest time to get a consultation	Before tests 8 and before assignment deadlines 6 and normal weeks 3 and always hard 1
Device they would use to book	Phone 7 and laptop 6
Would use an online booking system (1 to 5)	Average 4.2 and 9 of 13 gave 4 or 5

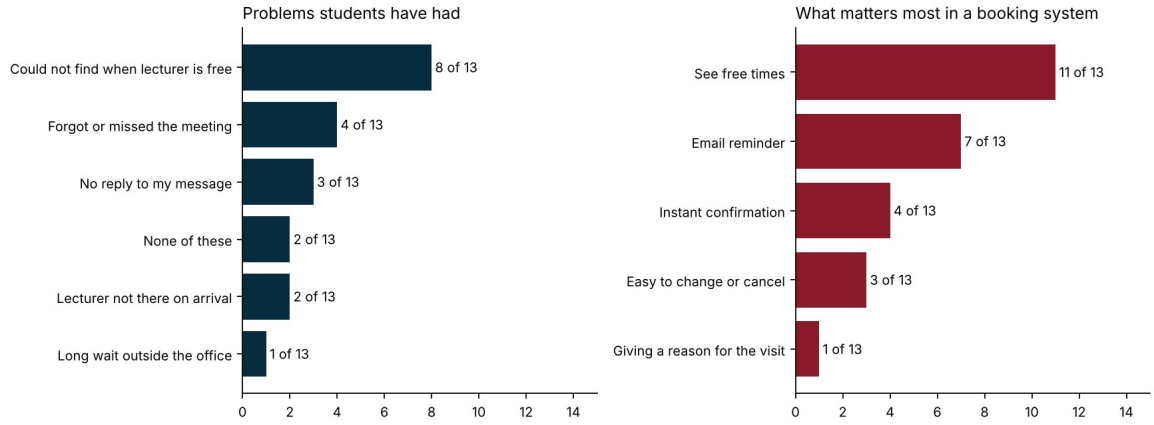


Figure 5.7 *Student survey results*

What the survey means for CampusBook

- The biggest problem is not knowing when a lecturer is free (8 of 13) and the most wanted feature is seeing free times (11 of 13) This is what the Book a slot page does
- Email reminders were chosen by 7 of 13 and 4 students had forgotten or missed a meeting so the reminder job (FR-08) answers a real need
- Instant confirmation was chosen by 4 and changing or cancelling by 3 which supports the confirmation email and the Change time and Cancel booking buttons
- Phone and laptop were almost equal (7 and 6) which supports testing both at 375 and 1366 pixels wide (NFR-03)
- Pressure is highest before tests and assignment deadlines so the busy weeks before tests are when CampusBook matters most

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The survey has limits It has 13 answers from one campus so it shows what these students said and not all students No lecturers answered it so the lecturer view comes from the user testing below

Table 5.10 *Feedback sources*

Feedback source	What it shows	When
Student survey (13 answers)	Current problems and what students want	22 October 2026
User testing with students and a lecturer	Time to complete a booking and problems seen and comments	28 October 2026
Supervisor feedback on this chapter	Corrections to the tests and the chapter	After the Week 4 submission

User Testing

The user testing was done on 28 October 2026 with 4 students and 2 lecturers who used CampusBook on their own and then answered a Google Form The target was 5 students and 1 lecturer so the lecturer target was met and the student group was one short Each tester is recorded only as S1 to S4 or L1 and L2 Students were asked to log in find an open slot book it and reach the confirmation and lecturers were asked to create or manage their slots Each tester rated how easy the task was from 1 (very difficult) to 5 (very easy) and how likely they are to use CampusBook instead of Teams or WhatsApp from 1 (very unlikely) to 5 (very likely) The bookings were not timed

Table 5.11 *User testing results*

Tester	Role	Task done	Task ease (1 to 5)	Would use it (1 to 5)	Suggestion
S1	Student	Booked and the confirmation was clear	5	4	Nothing
S2	Student	Booked and the confirmation was clear	5	3	Nothing
S3	Student	Booked and the confirmation was clear	4	5	A weekly view of all upcoming consultations and a notification when a lecturer adds new times
S4	Student	Booked and the confirmation was clear	4	3	Improve the look and make the navigation clearer
L1	Lecturer	Created and managed slots	3	2	Add a chat or call feature
L2	Lecturer	Created and managed slots	2	5	More features
Average		6 of 6 completed the task	3.83	3.67	

4 of the 6 testers (66.7%) rated the task easy or very easy which is below the 80% target The four students were positive with an average of 4.50 out of 5 and all four said it was easy to find a slot and understand the details and that the booking confirmation was clear The two lecturers could create and manage slots but rated the task 3 and 2 (average 2.50) so the Lecturer slots page needs clearer navigation and should be checked by watching lecturers use it

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Willingness to use CampusBook instead of Teams or WhatsApp averaged 3.67 out of 5 and 3 of the 6 testers gave 4 or 5 The suggestions were a weekly view of upcoming consultations with a notification when a lecturer adds new times clearer navigation a chat or call feature and more features in general Two testers suggested no change

5.7 Evaluation Against Requirements

Table 5.12 compares the app at commit e9ab578 with each requirement from Chapter 3 This is the result after the fixes Met means the tests passed and Partially Met means part of the requirement still needs work

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Table 5.12 *Requirements evaluation*

ID	Requirement	Result	Notes
FR-01	Students and lecturers log in with their email and password	Met	Login works for both roles and bad details give one general error
FR-02	The system checks the user role so students and lecturers only reach their own pages and actions	Met	Role checks on every route and page
FR-03	Lecturers publish edit and remove their own future consultation slots	Met	API and page work. Slots are published and edited and removed from the page after the D-03 and D-10 fixes
FR-04	Students view the open slots of a chosen lecturer	Met	Slots listed correctly and the times show after the D-02 fix
FR-05	Students book an open slot and give a short reason for the visit	Met	Booking saved and confirmation shown
FR-06	Students change or cancel their own upcoming bookings	Met	Cancel and change both work in the page after the D-04 fix
FR-07	The system stops two students booking the same slot	Met	8 simultaneous requests gave one booking
FR-08	The system sends an email confirmation after booking and a reminder before the meeting	Met	Confirmation and reminder work in preview mode and an SMTP test email was received on 23 October The booking emails also arrived on the demonstration laptop
FR-09	Students see their own booking history and lecturers see the bookings for their slots	Met	Student page and lecturer page both show the bookings after the D-03 fix
NFR-01	A first-time student completes a booking in under 2 minutes	Partially Met	Scripted journey took 1.8 seconds The bookings were not timed in the user testing on 28 October
NFR-02	The main pages load within 3 seconds in the team test set up	Met	Slowest page load was 389 ms and slowest API call 19 ms
NFR-03	Every page works on a phone at 375 pixels wide and on a laptop at 1366 pixels wide	Met	No sideways scrolling on all five pages at both widths
NFR-04	Passwords are stored as bcrypt hashes and never as plain text	Met	All 7 stored passwords are bcrypt hashes and repeated wrong passwords are limited after the D-06 fix
NFR-05	Each user can only see their own bookings or the bookings for their own slots	Met	Other users records were refused in every test
NFR-06	Only the data needed for logging in and booking is stored with made-up data during development	Met	Five tables and no sensitive columns and all users on the test domain
NFR-07	Errors explain what to fix and the form keeps what the user already typed	Met	Plain messages and typed text kept and a clear message when the session ends after the D-05 and D-08 fixes

9 of the 9 functional requirements and 6 of the 7 non-functional requirements are fully met and none are unmet

The only requirement that is not fully met is NFR-01 because the bookings were not timed in the user testing All four students booked a consultation and said the confirmation was clear but there is no time to compare with the 2 minute target FR-08 was tested with the email preview and an SMTP test email was sent and received on 23 October (Chapter 4 section 4.7) and the booking emails arrived on the demonstration laptop

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5.8 Conclusion

The tests show that the booking journey works from login to confirmation on a phone and a laptop and that two students can never hold the same slot 35 of 36 tests pass after the fixes and the one test not run is the timed booking test

Testing found real problems at the joins between the pages and the API The Lecturer slots page and the Book a slot page did not read the Slots API correctly and My bookings had no change button These and the other defects were fixed and each fix was retested and the fixed code was merged into the main branch

The student survey (13 answers) supports the need for the system and the features chosen and in the user testing 4 of the 6 users rated CampusBook easy to use with the students more positive than the lecturers

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Chapter 6 Conclusion and Recommendations

6.1 Introduction

This chapter concludes the CampusBook project, which was proposed to address the difficulty of arranging lecturer consultations through separate channels such as email, Microsoft Teams, WhatsApp and walk-in requests. When availability and requests are scattered across these channels, students may wait for responses or be uncertain whether a meeting has been confirmed.

Lecturers may also have difficulty maintaining a consistent view of upcoming consultations and the reasons students need assistance. The aim of CampusBook is to design, develop and evaluate a web-based consultation booking system that enables students to book, change and cancel consultation appointments while lecturers manage their available time slots. The project focuses on one CTU campus and a browser-based application. It does not include live integration with CTU systems, SMS notifications, video calls or a native mobile application.

6.2 Summary of the Study

The project identified a practical need for a central place where students can view available consultation slots and manage their bookings without relying on long message exchanges. The literature review considered Microsoft Bookings, Calendly and Moodle Scheduler.

These systems demonstrate that online scheduling can organise availability and appointments, but their fit depends on institutional requirements and the surrounding learning or business environment. CampusBook focuses on a defined campus use case rather than attempting to replace a full institutional timetable or learning-management system.

The system was built and tested in Week 4. All eight coding tasks were merged into the main branch, and 36 test cases were written and linked to the requirements. After the fixes, 35 of the 36 tests passed. The one test not run is the timed booking test, because the bookings were not timed in the user testing on 28 October. Testing logged 10 defects and all 10 were fixed and retested. Eight booking requests sent for one slot at the same moment produced one booking, which shows that two students cannot hold the same slot. The pages loaded in under 400 ms in the team test set-up against a target of 3 seconds.

The group also learned what worked and what did not. The shared page template, the pull request reviews and the task sign-up sheet worked well. The Lecturer slots page was merged late and did not

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match the Slots API, because the page and the API used different field names, and this was only found when the merged code was tested. In future the group would agree the API field names before coding and test every merged page against the API.

6.3 Achievement of Objectives

The table below compares the five objectives from Chapter 1 with the evidence from Chapters 4 and 5.

Table 6.1 *Achievement of objectives*

Objective	Result	Evidence
1 Study how consultations are arranged now with at least 10 students	Achieved	13 students answered the survey on 22 October so the target was met (Chapter 5 section 5.6)
2 Compare at least three existing scheduling systems	Achieved	Microsoft Bookings, Calendly and Moodle Scheduler were compared on features, strengths and limitations (Chapter 2)
3 Design and develop a web system with all six must-have features on a live database	Achieved	All eight coding tasks were merged into main and all 9 functional requirements were met in testing (Chapters 4 and 5)
4 Test the system so every must-have test case passes, a booking takes under 2 minutes and pages load within 3 seconds	Mostly achieved	35 of 36 tests passed after the fixes and the pages loaded in under 400 ms. The bookings were not timed in the user testing, so the 2 minute target is not confirmed (Chapter 5)
5 Evaluate the system with at least 5 students and 1 lecturer with 80% or more rating it easy to use	Partly achieved	6 users tested (4 students and 2 lecturers) and 4 of the 6 (66.7%) rated CampusBook easy to use against a target of 80%. Students averaged 4.50 and lecturers 2.50 out of 5 (Chapter 5 section 5.6)

So three of the five objectives were achieved, the fourth was mostly achieved and the fifth was partly achieved.

6.4 Answers to Research Questions

How do students and lecturers at CTU arrange consultations now and what problems do they face?

Among 13 respondents, Teams was the most selected current method (11/13, 84.6%), followed by asking in class (5, 38.5%), office visits (3, 23.1%) and WhatsApp (2, 15.4%); none selected email. Unclear lecturer availability was the leading problem (8/13, 61.5%), followed by missed meetings (4, 30.8%) and no reply (3, 23.1%). Ten respondents (76.9%) reported a one-to-two-day wait for confirmation, and seven (53.8%) had given up seeking a consultation.

What do existing scheduling systems offer and what do they lack for a campus setting?

Microsoft Bookings, Calendly and Moodle Scheduler all let someone publish time slots and let others book them, so basic booking is already well covered. Their weakness is fit and cost for a campus.

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Microsoft Bookings is built for businesses, has no concept of students or modules, and needs administrator set-up. Calendly charges per lecturer for reminders and each lecturer gives a separate link, with the data held by a third party. None of them gives CTU students one free place to book any lecturer, which is the gap CampusBook was built to fill.

How can a web system let students book consultations and lecturers manage their time?

CampusBook does this with a React front end, a Node.js and Express API, and a MySQL database. Users log in with a role. Lecturers publish, edit and remove their own future slots, and students see a lecturer's open slots and book one with a reason for the visit. The server enforces the rules: slots cannot be in the past or overlap, and a slot that is already booked cannot be edited or removed by the lecturer. A database row lock and a unique key on the active booking stop two students taking the same slot. Confirmation and reminder emails are sent with Nodemailer, and both roles can see their booking history.

Does the system meet its functional and non-functional requirements?

Mostly yes. After the fixes, all 9 functional requirements and 6 of the 7 non-functional requirements were fully met and none were unmet. NFR-01 is only partly met: the scripted booking journey took 1.8 seconds, but the bookings were not timed in the user testing, so the two-minute target for first-time students is not confirmed. Testing found 6 failures and one blocked test on the merged code, and the 10 defects logged were all fixed and retested. Passwords are stored as bcrypt hashes, wrong roles are refused with 403 and missing tokens with 401, and repeated wrong passwords are limited. Confirmation and reminder emails were tested in preview mode, and an SMTP test email was received on 23 October (Chapter 4 section 4.7), and the booking emails arrived on the demonstration laptop.

Do students and lecturers find the system easy to use and useful?

Partly. The user testing on 28 October with 4 students and 2 lecturers answers this question. 4 of the 6 users (66.7%) rated CampusBook easy or very easy to use, which is below the 80% target. All four students found it easy to find a slot, said the booking confirmation was clear and rated the task 4.50 out of 5 on average. Both lecturers could create and manage slots but rated the task 3 and 2, so the lecturer page needs clearer navigation. Willingness to use CampusBook instead of Teams or WhatsApp averaged 3.67 out of 5. The survey adds stated interest from a wider group. Nine of 13 respondents (69.2%) gave a likelihood-to-use rating of 4 or 5 out of 5, and eight (61.5%) gave a rating of 5. Seeing a lecturer's free times was the most selected desired feature (11/13, 84.6%), followed by an email

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reminder (7/13, 53.8%). These findings suggest that respondents see potential value in the proposed system, and the booking journey itself needs few steps.

6.5 Limitations

Limited institutional scope: CampusBook was designed for one CTU campus and does not establish whether the same workflow will suit other campuses or departments.

Small sample: The survey had 13 student answers from one campus and no lecturer answered it, so it shows what these students said and not what all students and lecturers think. The user testing had only six participants, 4 students and 2 lecturers, which is one student short of the target, and the bookings were not timed.

Time limit: The system was built and tested in about two weeks by five members, so the user testing was short and was done in the final week.

Test environment: The performance times were measured with the API and the database on the same machine, so they leave out the network delay of a real campus connection.

Email dependency: Confirmation and reminder delivery depends on SMTP configuration and external email service availability. Messages may be delayed, rejected or classified as spam.

Scope exclusions: There is no admin dashboard, no SMS, no video calls and no link to CTU student systems because these were outside the scope set in Chapter 1.

6.6 Recommendations and Future Work

Keep testing on the main branch. Run the full set of tests again whenever new work is merged, so that the version on GitHub always matches the tested system.

Improve email observability and recovery. Record the outcome for each recipient separately, log SMTP failures without exposing passwords or message secrets, and implement a controlled retry process. This will help distinguish a student email success from a lecturer email failure.

Make usability changes based on evidence. Prioritise features supported by the actual survey: clear visibility of available times (selected by 11 of 13 respondents), reliable confirmation and email reminders (selected by 7 of 13), and a straightforward way to change or cancel a booking (selected by 3 of 13). Then validate the changes in a separate prototype usability evaluation.

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Improve the lecturer page. The two lecturers rated their task 3 and 2 out of 5 while the students averaged 4.50, so the Lecturer slots page needs clearer navigation. Watch lecturers use it and fix what slows them down.

Add automated tests. Run the test cases automatically on every pull request so that a page that does not match the API is caught before it is merged.

Run a pilot. Give one CTU department a pilot and collect real booking numbers before wider use.

Future Work

Additional notification options, such as SMS or other approved channels, while retaining a reliable email fallback.

Optional online meeting links for consultations that do not require in-person attendance. Video calls were excluded from the current scope.

Links to CTU student systems and calendar synchronisation, so that bookings appear in a lecturer's or student's own calendar.

A weekly calendar view of upcoming consultations and a notification when a lecturer adds new free times, both asked for in the user testing.

An admin view so that CTU staff can see bookings across lecturers and manage users.

A hosted version on a real server with HTTPS and a real email service.

6.7 Conclusion

CampusBook was developed in response to a coordination problem: consultation requests can be spread across several communication channels, making lecturer availability, booking confirmation and appointment history difficult to manage consistently. The 13 Google Forms responses provide preliminary support for this problem: 8 respondents (61.5%) could not find out when the lecturer was free, and 7 (53.8%) said they had given up on getting a consultation because of these types of problems. A focused web-based workflow could bring these activities together by allowing lecturers to publish slots and students to create and manage bookings in one place.

The project went on to build that workflow. Students and lecturers log in, lecturers publish slots, students book with a reason, and both receive email confirmations and reminders. In testing, 35 of 36 tests passed after the fixes and two students could never hold the same slot. In the user testing with 4 students

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and 2 lecturers, all four students rated the system easy to use, while the lower lecturer ratings show that the lecturer page is the next thing to improve. Overall, CampusBook offers a scoped and tested foundation for lecturer consultation booking.

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Appendices

The appendices hold the supporting documents for this dissertation

- Appendix A Test Plan
- Appendix B Ethical Clearance and Consent
- Appendix C Test Cases and Defect Log
- Appendix D Project Schedule and Risk Register
- Appendix E Group Declaration and Contribution Forms
- Appendix F Attendance Register
- Appendix G Demo Script
- Appendix H Presentation Slides

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Appendix A Test Plan

1. Test Objectives

The test plan shows that CampusBook meets the nine functional and seven non-functional requirements in the SRS and that the must have features work together on a phone and a laptop

- Show that a student can log in book change and cancel a consultation and that a lecturer can manage slots
- Show that two students can never hold the same slot even when they book at the same moment
- Show that users only reach their own data and that passwords are stored safely
- Show that the main pages load in under 3 seconds and work at 375 and 1366 pixels wide
- Find defects early and record each one in the Defect Log until it is fixed and retested
- Show that every requirement is linked to at least one test in the Requirements Traceability Matrix

2. Scope

Included

- Back-end API routes for login slots and bookings (routes/auth.js slots.js and bookings.js)
- Role and token checks (middleware/auth.js) and input checks (middleware/validate.js)
- Confirmation and reminder emails (services/mailer.js and jobs/reminders.js)
- Database rules in schema.sql including the unique key that blocks double booking
- Front-end pages Login and Book a slot and My bookings and the Lecturer slots page and route protection
- Phone and laptop layouts and page load times

Not included in this round

- Real email delivery through an SMTP service (emails are checked in the preview mode)
- Timed bookings with real students (the user testing on 28 October with 4 students and 2 lecturers collected ratings and comments)
- Load testing with many users and testing on a real phone (the phone size is tested in the browser)

3. Test Environment

Item	Detail
Application version	Branch main at commit e9ab578 on 22 October 2026 and again after the fixes for D-02 to D-10
Back end	Node.js 22 with Express and mysql2 on port 4100
Database	MariaDB 10.11 (MySQL compatible) built from database/schema.sql and seed.sql
Front end	React 18 and Vite 5 production build served by vite preview
Browser and sizes	Chromium driven by Playwright at 375 x 800 (phone) and 1366 x 768 (laptop)
Test data	Seven made up users on the campusbook.test domain and five seeded slots
Email	Preview mode with no SMTP host so emails print to the server log
Test scripts	test_scripts/api_tests.js and test_scripts/ui_tests.js in folder 06

The two scripts are kept in folder 06 as the record of what was run

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4. Test Types

Test type	What it covers	Tool
Unit and API	Every route with valid and invalid input and wrong roles	Node script calling the API
Integration	API with database and emails and the reminder job	Node script and SQL checks
System and UI	Full journeys in the browser and route protection	Playwright with Chromium
Security	Role checks and forged tokens and access to other records and password storage	Node script and SQL checks
Performance	Page load times and API response times	Browser and script timings
Usability	Phone and laptop layouts and error messages and a scripted journey	Playwright and a heuristic check
Acceptance	User testing with real students and a lecturer	Done on 28 October with a Google Form (bookings not timed)

5. Entry and Exit Criteria

Entry criteria

- Code for the feature is merged into main
- The database builds cleanly from the two SQL scripts
- The API starts and the health check answers

Exit criteria

- Every Must requirement has at least one test with an actual result
- Every failed test has a defect in the Defect Log
- No open defect stops the main booking journey
- Every defect is fixed and retested before the final submission

6. Roles and Responsibilities

Member	Testing role
Willem Gerber	Test lead and Group Lead. Wrote the Test Plan and test cases and ran the tests and kept the Defect Log and the Traceability Matrix
Oscar Khumalo	Checks and retests fixes for the database connection and the email and reminder services
Lebohang Tlaitlai	Checks and retests fixes for the Slots API and the Login page and the student survey
Thomas du Plessis	Builds and retests the Lecturer slots page
Gerhardus van den Berg	Checks the schema and the test data

7. Defect Management

Each failed test becomes a row in the Defect Log with a date and the feature and a description and a severity and a priority and the owner

Severity	Meaning
Critical	The app stops or data is lost or another users data is exposed
High	A Must feature does not work or is missing
Medium	A feature works but with a clear problem or a missing part
Low	A small usability or wording problem

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The developer who owns the file fixes the defect then the test is run again and the result is written in the Retest Result column before the defect is closed

Priority follows the demo plan so anything that breaks the booking journey is fixed first

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Appendix B Ethical Clearance and Consent

Project Title

CampusBook: a lecturer consultation booking system for CTU Training Solutions

Data / Participants

Activity	Involved	Details
Personal data	Yes	Names emails and bookings with only made-up data during development
Interviews	No	No formal interviews and our lecturer reviewed the proposal
Survey	Yes	Anonymous student survey done on 22 October 2026 and answered by 13 CTU students with no names or student numbers
User testing	Yes	4 students and 2 lecturers on 28 October with a short anonymous Google Form

Ethical Considerations

- Consent: every participant gets a short information sheet and takes part voluntarily and can stop at any time
  User testing participants sign a consent form and survey participants tick a consent box before starting
- Privacy and confidentiality: survey answers are anonymous and testing notes use codes such as S1 instead of names
  Results are only reported as totals or anonymous quotes
- Secure data handling: research data is kept in a private password-protected folder and deleted once the project is marked
- Data minimisation: the system only collects what it needs and hashes passwords and shows users only their own bookings in line with POPIA

Confirmation

☐ No human participant data required ☒ Ethical considerations reviewed ☒ Additional approval required

Human participant data was collected in the anonymous student survey (13 answers) and again in the user testing on 28 October (4 students and 2 lecturers) and the supervisor approved both before they started

Supervisor Sign-off

Supervisor: Jabulile Mashibini Signature: _____ Date: _____

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Appendix C Test Cases and Defect Log

Test cases and results

Test ID	Feature	Expected Result	Status
TC-01	Login	Valid logins return a token and role. Wrong password and unknown email give the same general error	Pass
TC-02	Role checks on the server	Wrong role gets 403 and no token gets 401	Pass
TC-03	Lecturer manages own slots	Owner can publish edit and remove. Another lecturer gets 404	Pass
TC-04	Student views open slots	Only future slots with status open are listed	Pass
TC-05	Book a slot	Booking saved with student slot and reason and the slot becomes booked	Pass
TC-06	Change and cancel a booking	Old slot opens after a change and after a cancel. Second cancel is refused	Pass
TC-07	No double booking	Exactly one confirmed booking and the rest get the slot taken message	Pass
TC-08a	Confirmation email	Confirmation to the student and a notice to the lecturer and the confirmation is marked sent	Pass
TC-08b	Reminder email	Reminder sent to student and lecturer and marked sent	Pass
TC-09	Booking history	Each user sees only their own records and cancelled bookings stay in the history	Pass
TC-10	First time student books in under 2 minutes	Under 120 seconds with real students	Not Run
TC-11	Main pages load in under 3 seconds	Each page loads in under 3000 ms	Pass
TC-12	Phone and laptop layout	No sideways scrolling on any page	Pass
TC-13	Passwords stored as bcrypt hashes	Only 60 character bcrypt hashes and login uses bcrypt compare	Pass
TC-14	Users cannot reach other records	All refused with 404 and nothing changes	Pass
TC-15	Only needed data is stored	No sensitive columns and every user is on the test email domain	Pass
TC-16	Errors explain what to fix (API)	400 with a plain message and nothing saved	Pass
TC-16b	Errors on screen and kept input	Plain messages and the typed text is kept	Pass
TC-17	Book a missing past or taken slot	409 with Sorry that slot is no longer available	Pass
TC-18	Slot validation rules	Refused with a message that says what to fix	Pass
TC-19	Booked slot cannot be edited or removed	409 for both	Pass
TC-20	Removing a slot keeps history	Slot marked cancelled and history kept and publishing again reopens it	Pass
TC-21	Bad ids bad JSON and unknown routes	400 or 404 with a clear message and no server error	Pass
TC-22	Forged expired and garbage tokens	All refused with 401	Pass
TC-23	Health check	200 and ok true	Pass
TC-24	API response time for the slot list	Under 500 ms for every request	Pass
TC-25	Lecturer publishes a slot from the page	The page lists the lecturer slots and publishing saves the slot	Pass
TC-26	Reminder not sent for a cancelled booking	No reminder left to send	Pass
TC-27	Pages protected by login and role	Visitors go to login and each role is sent to their own home page	Pass
TC-28	Expired or invalid session	User is sent to the login page with a clear message	Pass
TC-29	Change a booking from the page	A change button is offered next to cancel and moving the booking works	Pass
TC-30	Repeated wrong passwords are limited	The attempts are blocked after several failures and other accounts are not affected	Pass

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Test ID	Feature	Expected Result	Status
TC-31	Database refuses a second confirmed booking	Insert fails with a duplicate key error	Pass
TC-32	Times shown on the Book a slot page	Each time button shows the start and end like 09:00 to 09:30	Pass
TC-33	Lecturer removes a slot from the page	The slot is removed and the list updates	Pass
TC-34	Lecturer edits a slot from the page	The page offers an edit option and the change is saved	Pass

Defect log

Defect ID	Feature	Description	Severity	Status	Retest Result
D-01	Bookings API and My bookings	Booking dates came back as time zone shifted dates because the raw slot_date was returned so a booking could show on the wrong day	High	Closed	Pass: dates match the stored dates (TC-05 and TC-09)
D-02	Book a slot page	Time buttons show only "to" and the confirm panel shows "undefined to undefined" because the page reads start and end but the Slots API sends startTime and endTime	High	Fixed	Pass: 09:00 to 09:30 and 09:30 to 10:00 (TC-32)
D-03	Lecturer slots page	The merged page does not work with the Slots API: it calls /api/slots/my but the route is /api/slots/mine so the list shows That API route does not exist and it sends start and end but the API needs startTime and endTime so publishing fails (FR-03 and FR-09)	High	Fixed	Pass: the list shows the slots and publish and remove both work (TC-25 and TC-33)
D-04	My bookings page	No change button although the API supports changing a booking (FR-06)	Medium	Fixed	Pass: the booking moved and the notice showed (TC-29)
D-05	Session handling	An expired or invalid token leaves the user on the page with an error and the old token is kept instead of going to the login page	Medium	Fixed	Pass (TC-28)
D-06	Login security	There is no limit on repeated wrong passwords so 40 attempts in a row were all answered	Medium	Fixed	Pass: attempts 11 and 12 gave 429 and another account still logged in (TC-30)
D-07	Documentation	Chapter 4 says the reminder scheduler runs every 15 minutes but the code runs it every minute	Low	Fixed	Pass: the text matches jobs/reminders.js
D-08	Book a slot page	After a failed booking the reason box is hidden until a time is chosen again although the typed text is kept	Low	Fixed	Pass (TC-16b)
D-10	Lecturer slots page	The page can publish and remove slots but has no way to edit a slot although the API supports editing (FR-03)	Medium	Fixed	Pass: the new location was saved (TC-34)
D-09	Route protection	A logged in user who opens the other roles page is sent to the login page and not to their own home page	Low	Fixed	Pass (TC-27)

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Appendix D Project Schedule and Risk Register

Project schedule and sprint backlog

Task ID	Task / User Story	Owner	Due Date	Status
T01	Chapter 1 Introduction	All members	19 Oct 2026	Complete
T02	Chapter 2 Literature review and matrix	All members	19 Oct 2026	Complete
T03	Chapter 3 Requirements and methodology	All members	19 Oct 2026	Complete
T04	Chapter 4 UML architecture and wireframes	All members	19 Oct 2026	Complete
T05	Database design and SQL scripts	Gerhardus van den Berg	19 Oct 2026	Complete
T06	Project charter ethics form and work plan	Willem Gerber	19 Oct 2026	Complete
T07	GitHub repository set up and first commits	All members	19 Oct 2026	Complete
T08	Back-end set up and login	Willem Gerber	20 Oct 2026	Complete
T09	Slots API	Lebohang Tlaitlai	22 Oct 2026	Complete
T10	Bookings API	Willem Gerber	21 Oct 2026	Complete
T11	Database and emails	Oscar Khumalo	21 Oct 2026	Complete
T12	Front-end set up and Login page	Lebohang Tlaitlai	21 Oct 2026	Complete
T13	Book a slot page	Willem Gerber	21 Oct 2026	Complete
T14	Lecturer slots page	Thomas du Plessis	22 Oct 2026	Complete
T15	My bookings page and phone layout	Willem Gerber	22 Oct 2026	Complete
T16	Student survey	Gerhardus van den Berg	22 Oct 2026	Complete
T17	Testing workbooks and test cases	Willem Gerber	23 Oct 2026	Complete
T18	Chapter 5 and Chapter 4 sections 4.6 to 4.8	All members	23 Oct 2026	Complete
T19	Demo script and demo video	All members	23 Oct 2026	Complete
T20	Back-end fixes and security check	Willem Gerber	26 Oct 2026	Complete
T21	Front-end fixes and polish	Lebohang Tlaitlai	26 Oct 2026	Complete
T22	User testing with 4 students and 2 lecturers	Oscar Khumalo	28 Oct 2026	Complete
T23	Final Chapter 5 and Chapter 6	All members	26 Oct 2026	Complete
T24	Demo data demo laptop and backup video	All members	27 Oct 2026	Complete
T25	Final slides and practice runs	All members	28 Oct 2026	Complete
T26	Final dissertation and submission	Willem Gerber	29 Oct 2026	Complete

Risk issue and change register

ID	Type	Description	Owner	Status
R-01	Risk	Not enough time to build all six features	Willem Gerber	Resolved
R-02	Risk	Work falls on one or two members	Willem Gerber	Resolved
R-03	Risk	Some members are new to React or Express	Oscar Khumalo and Thomas du Plessis	Resolved
R-04	Risk	Too few survey or test participants	Lebohang Tlaitlai	Resolved
R-05	Risk	Email reminders not working in time	Oscar Khumalo	Resolved
R-06	Risk	Double booking or one user seeing another user data	Oscar Khumalo	Resolved
I-01	Issue	Survey target was 10 students	Willem Gerber	Resolved
I-02	Issue	Supervisor approval for the survey and user testing	Willem Gerber	Resolved
I-03	Issue	First commits from every member	All members	Resolved
C-01	Change	Lecturer interviews replaced by lecturer feedback on the proposal (No formal interviews were held)	Willem Gerber	Resolved
C-02	Change	Survey target set to at least 10 students everywhere (Two different numbers were used)	Willem Gerber	Resolved
C-03	Change	App screenshots replaced by wireframes (The app is built in Week 4)	Willem Gerber	Resolved
C-04	Change	Test set up for the page load target written down (Makes NFR-02 repeatable)	Willem Gerber	Resolved

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Appendix E Group Declaration and Contribution Forms

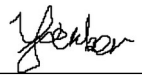
Academic and Contribution Declaration

We the members of SE2 Group 3 declare that CampusBook and the dissertation are our own work Every source we used is referenced in Harvard style and each member contribution is recorded in our Individual Contribution Forms and in our GitHub commits

Member Roles

Full name	Student number	Role	Main responsibilities
Willem Gerber	20230257	Group Lead and Scrum Master	Stand-ups, GitHub and reviews, back-end login and bookings, Book a slot and My bookings pages, and compiling the submissions
Oscar Khumalo	20230279	Back-end developer	Requirements database connection and email reminders Chapter 4.7 and 4.8 and Chapter 5
Thomas du Plessis	20230260	Front-end developer	Wireframes use case and activity diagrams Lecturer slots page Chapter 4.6 slides and demo video
Lebohang Tlaitlai	20233068	Testing and research lead	Literature review Chapter 3 and SRS student survey React set up Login page Slots API and Chapter 6
Gerhardus van den Berg	20230240	Database developer	ERD class diagram database design and SQL scripts (schema and test data) and presentation slides 5 to 7

Signatures

Name	Signature	Date
Willem Gerber		2026/10/29
Oscar Khumalo	*B.O Khumalo*	2026/10/29
Thomas du Plessis		2026/10/29
Lebohang Tlaitlai	*LN.Tlaitlai*	2026/10/29
Gerhardus van den Berg		2026/10/29

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Individual Contribution Form Willem

Student Details

Name: Willem Gerber Student No.: 20230257 Group: SE2 Group 3

Contribution Summary

Area	What I did
Research	Chapter 1 Introduction and help with the Chapter 2 literature review
Design	Architecture diagram and the shared page template
Coding	Back-end set up with login and role checks, the Bookings API, and the Book a slot and My bookings pages
Testing	Test Plan, test cases, Defect Log and fixes
Documentation	Project charter, ethics form and compiling the submissions and final dissertation
Project management	Group Lead and Scrum Master, GitHub set-up and pull request reviews
Presentation	Slides 1 to 4 in the final presentation (Week 5)

Git / Evidence References

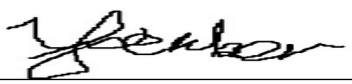
Commit or file	Date	What it shows
dcb7cd2	15 Oct	Repository set up
6f458c7 and 590e891	20 and 21 Oct	Login, Bookings API and Book a slot page
87cefdb	22 Oct	My bookings page

Peer Confirmation

Member	Accurate (Yes or No)	Signature	Date
Thomas du Plessis	Yes		2026/10/29
Oscar Khumalo	Yes	*B.O Khumalo*	2026/10/29
Lebohang Tlaitlai	Yes	*LN.Tlaitlai*	2026/10/29
Gerhardus van den Berg	Yes		2026/10/29

Student Declaration

I confirm that the information above accurately reflects my contribution

Name	Signature	Date
Willem Gerber		2026/10/29

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Individual Contribution Form Oscar

Student Details

Name: Oscar Khumalo Student No.: 20230279 Group: SE2 Group 3

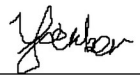
Contribution Summary

Area	What I did
Research	Functional and non-functional requirements for Chapter 3 and the SRS
Design	Sequence diagram
Coding	Database connection (db.js) plus the confirmation and reminder emails (mailer.js and reminders.js)
Documentation	Chapter 4.7 Working System (key modules) and 4.8 Version Control with screenshots (Week 4) and the final Chapter 5 (Week 5)
Testing	User testing with 4 students and 2 lecturers on 28 October using a Google Form and the user testing results for Chapter 5 (Week 5)
Presentation	Slides 11 to 13 in the final presentation (Week 5)

Git / Evidence References

Commit or file	Date	What it shows
5682abf	21 Oct	Database and email reminder services
Pull request #3	21 Oct	Merged into main
docs/diagrams/sequence.png	Week 3	Sequence diagram

Peer Confirmation

Member	Accurate (Yes or No)	Signature	Date
Willem Gerber	Yes		2026/10/29
Thomas du Plessis	Yes		2026/10/29
Lebohang Tlaitlai	Yes	*LN.Tlaitlai*	2026/10/29
Gerhardus van den Berg	Yes		2026/10/29

Student Declaration

I confirm that the information above accurately reflects my contribution

Name	Signature	Date
Oscar Khumalo	B.O Khumalo	2026/10/29

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Individual Contribution Form Thomas

Student Details

Name: Thomas du Plessis Student No.: 20230260 Group: SE2 Group 3

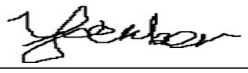
Contribution Summary

Area	What I did
Design	Wireframes plus the use case and activity diagrams
Coding	Lecturer slots page (commit c2ecca1)
Documentation	Chapter 4.6 Implementation (development environment)
Presentation	First slide deck, the Demo Script and the demo and backup video (Week 4) and slides 14, 15 and 17 in the final presentation (Week 5)

Git / Evidence References

Commit or file	Date	What it shows
docs/wireframes	Week 3	Login Book a slot Lecturer and phone wireframes
docs/diagrams/usecase.png and activity.png	Week 3	Use case and activity diagrams
c2ecca1	22 Oct	Add lecturer consultation slots page
Pull request #6	22 Oct	Reviewed and merged

Peer Confirmation

Member	Accurate (Yes or No)	Signature	Date
Willem Gerber	Yes		2026/10/29
Oscar Khumalo	Yes	*B.O Khumalo*	2026/10/29
Lebohang Tlaitlai	Yes	*LN.Tlaitlai*	2026/10/29
Gerhardus van den Berg	Yes		2026/10/29

Student Declaration

I confirm that the information above accurately reflects my contribution

Name	Signature	Date
Thomas du Plessis		2026/10/29

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Individual Contribution Form Lebohang

Student Details

Name: Lebohang Tlaitlai Student No.: 20233068 Group: SE2 Group 3

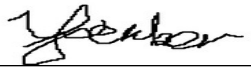
Contribution Summary

Area	What I did
Research	Literature review and the student survey
Requirements	Chapter 3 and the SRS
Coding	React app set up page routes and the Login page Slots API for lecturers and students (routes/slots.js)
Documentation	Chapter 6 Conclusion and Recommendations
Presentation	Slides 8 to 10 in the final presentation and driving the app in the live demo (Week 5)

Git / Evidence References

Commit or file	Date	What it shows
fe19ea8	21 Oct	Set up React app routes and Login page
e9b5f3e	22 Oct	Add Slots API routes
Pull requests #2 and #4	21 and 22 Oct	Merged into main

Peer Confirmation

Member	Accurate (Yes or No)	Signature	Date
Willem Gerber	Yes		2026/10/29
Oscar Khumalo	Yes	*B.O Khumalo*	2026/10/29
Thomas du Plessis	Yes		2026/10/29
Gerhardus van den Berg	Yes		2026/10/29

Student Declaration

I confirm that the information above accurately reflects my contribution

Name	Signature	Date
Lebohang Tlaitlai	LN.Tlaitlai	2026/10/29

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Individual Contribution Form Gerhardus

Student Details

Name: Gerhardus van den Berg Student No.: 20230240 Group: SE2 Group 3

Contribution Summary

Area	What I did
Design	ERD and class diagram used in Chapter 4.3 UML Design and 4.4 Database Design
Database	Database design and SQL scripts (task T05): MySQL schema (schema.sql) and the test data (seed.sql)
Documentation	Team work on Chapters 1 to 4 (tasks T01 to T04)
Testing	The seed.sql test data used for the 36 test cases and the live demo
Presentation	Slides 5 to 7 in the final presentation (Week 5)

Git / Evidence References

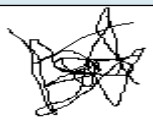
Commit or file	Date	What it shows
docs/diagrams/erd.png and class.png	Week 3	ERD and class diagram
database/schema.sql and seed.sql	Week 3	Schema and test data

Peer Confirmation

Member	Accurate (Yes or No)	Signature	Date
Willem Gerber	Yes		2026/10/29
Oscar Khumalo	Yes	*B.O Khumalo*	2026/10/29
Thomas du Plessis	Yes		2026/10/29
Lebohang Tlaitlai	Yes	*LN.Tlaitlai*	2026/10/29

Student Declaration

I confirm that the information above accurately reflects my contribution

Name	Signature	Date
Gerhardus van den Berg		2026/10/29

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Appendix F Attendance Register

Project attendance register

Date	Full Name	Task Completed
15 Oct 2026	Willem Gerber	Repository and project structure (commits dcb7cd2 and 092dd77)
19 Oct 2026	Willem Gerber	Back-end folders schema seed scripts and diagrams (commit 88460f8)
20 Oct 2026	Willem Gerber	Header layout theme login middleware and Bookings API (commit 6f458c7 and pull request #1)
21 Oct 2026	Lebohang Tlaitlai	React app routes and Login page (commit fe19ea8 and pull request #2)
21 Oct 2026	Willem Gerber	Booking error handling routes and Book a slot page (commit 590e891)
21 Oct 2026	Oscar Khumalo	Database and email reminder services (commit 5682abf)
21 Oct 2026	Willem Gerber	Reviewed and merged pull request #3
22 Oct 2026	Lebohang Tlaitlai	Slots API routes (commit e9b5f3e and pull request #4 merged at 07:50)
22 Oct 2026	Willem Gerber	My bookings page and phone layout (commit 87cefdb)
22 Oct 2026	Thomas du Plessis	Lecturer slots page (commit c2ecca1 and pull request #6 merged at 14:56)
22 Oct 2026	Willem Gerber	Reviewed and merged pull request #6
23 Oct 2026	Oscar Khumalo	Improved booking and reminder email status handling (commit 5aff8cb)
23 Oct 2026	Oscar Khumalo	Merged main into the Oscar branch (commit bfc98b0)
24 Oct 2026	Lebohang Tlaitlai	Frontend login page fixed and Backend polishes (commit a72413d and pull request #7)

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Appendix G Demo Script

1. Opening

Good day. We are Group 3 and our project is CampusBook, a consultation booking system for CTU Training Solutions.

Our members are Willem Gerber, Oscar Khumalo, Thomas du Plessis, Lebohang Tlaitlai and Gerhardus van den Berg.

Today we explain the problem, show the student and lecturer journeys live and share our testing evidence.

2. Problem and Objectives

At CTU, consultations are arranged through Teams, email, WhatsApp or by visiting a lecturer. Students do not know when a lecturer is free and lecturers struggle to organise appointments.

We surveyed 13 students. 11 needed a consultation in the last 3 months, 7 gave up because of the problems and 9 said they would use an online booking system.

CampusBook gives one place where lecturers publish consultation slots and students book, change or cancel them.

Our objectives were to study how consultations work now, compare existing systems, build the essential features, test performance and security, and evaluate ease of use with students and lecturers.

3. Architecture / Design

CampusBook has three tiers. The React frontend uses Vite and React Router. The Node.js and Express backend handles login, slots, bookings and reminders. A MySQL database stores users, lecturer profiles, slots, bookings and notifications.

Logins use JWT tokens and passwords are stored as bcrypt hashes. The system checks the role on every route so students and lecturers only reach their own pages. Nodemailer sends confirmation emails and node-cron sends the reminders.

Show the architecture diagram and then the ERD and point out how users, slots and bookings link together.

4. Live Demonstration Flow

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Follow these steps in order. Keep each step to about 30 seconds.

Step	What to do	What the audience should see
1	Open CampusBook and log in as a lecturer	The lecturer lands on My Consultation Slots
2	Publish a slot with a future date, start and end time and a location	A success message and the new slot in the list
3	Edit an available slot	The updated slot. Explain that booked slots cannot be edited or removed
4	Log out and log in as a student, then open Book a Consultation	The lecturer list and the available dates
5	Pick a lecturer, date and time, enter a reason and confirm	A confirmation with lecturer, date, time and location
6	Open My Bookings, then change or cancel the booking	The booking list and slot availability update
7	Show the phone-sized layout and a wrong password or expired session message	The page fits the phone and the message explains what to do

If there is time, explain test TC-07: 8 simultaneous requests for one slot gave 1 booking and 7 refusals. Do not run the load test live.

5. Testing Evidence

We wrote 36 test cases. 35 passed, none failed and one was not run: the timed test of a first-time student booking, because the bookings were not timed in the user testing.

Testing found 10 defects. All were fixed and retested. Two important ones were a wrong route on the lecturer slots page (/api/slots/mine) and consultation times showing as "undefined". We also fixed rescheduling, expired sessions and login rate limiting.

Security tests showed that wrong roles are refused, passwords are bcrypt hashes and a slot cannot be booked twice. The slowest page loaded in 389 ms against a 3 second target and the pages work on phone and laptop sizes.

The survey result shows that students want the system. In the user testing on 28 October with 4 students and 2 lecturers, 4 of the 6 rated CampusBook easy to use.

6. Conclusion

CampusBook brings consultation booking and lecturer availability into one web application. Login, slot management, booking, reminders and the phone layout all work and were tested.

Future ideas are links to CTU student systems and calendar synchronisation. Thank you. We are happy to take questions.

7. Backup Plan

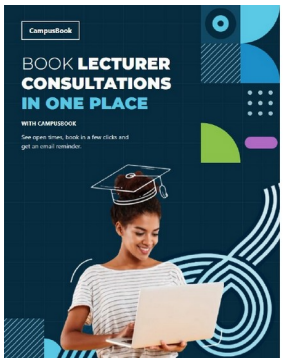
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Keep a 3 to 5 minute screen recording ready. It shows publishing a slot, student login, a booking, My Bookings and one error message. The group walkthrough video can be used for this.

- Keep screenshots and the test data on the laptop
- Start the database, backend and frontend before the presentation and test them
- If the internet or a service fails, play the recording and use the screenshots
- Tell the audience it is a recording and do not say it is live
- Keep passwords and JWT secrets out of screenshots and the recording

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Appendix H Presentation Slides



SPM622 class presentation

CampusBook

A web app for booking lecturer consultations at CTU

SE2 Group 3

Willem Gerber, Oscar, Thomas, Lebohang and Gerhardus
CTU Training Solutions | 29 October 2026

1

THE PROBLEM

Booking a consultation **is messy today**

How students ask today

- Teams message
- Asking in class
- Office visit
- WhatsApp

What goes wrong

- Nobody can see when a lecturer is free
- Students wait days for a reply
- Meetings are forgotten, missed or double-booked
- Lecturers do not know who is coming or why

It is worst before tests and assignment deadlines, exactly when students need help most.

CampusBook | SE2 Group 3 | CTU Training Solutions 2

THE EVIDENCE

We asked 13 students. **This is what they said**

8 of 13
cannot tell when a lecturer is free

10 of 13
wait 1 to 2 days to get a confirmed time

7 of 13
have given up on getting a consultation

Seven out of thirteen gave up. Students are going without help because asking is too hard.

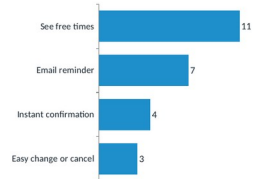
Source: Group 3 student survey, 13 anonymous answers, 22 October 2026

3

THE EVIDENCE

The survey told us **what to build**

What students want most (out of 13)



So CampusBook must:

- Show each lecturer's open times
- Confirm the booking straight away
- Send a reminder before the meeting
- Let students change or cancel

4.2 out of 5 average for "Would you use an online booking system?" 9 of 13 gave 4 or 5.

CampusBook | SE2 Group 3 | CTU Training Solutions 4

THE PLAN

Our aim and **five objectives**

OUR AIM

Build and test a web system where students book, change and cancel consultations, and lecturers manage the time they are free.

1. **Find out how consultations work now** — A survey of students
2. **Compare existing booking tools** — Microsoft Bookings, Calendly and Moodle
3. **Build the six must-have features** — Working on a live database
4. **Test it** — Check that every feature works, fast, on a phone and a laptop
5. **Evaluate it with real users** — User testing with students and lecturers

CampusBook | SE2 Group 3 | CTU Training Solutions 5

THE RESEARCH

Why not use a tool **that already exists?**

Microsoft Bookings
Good: syncs with Outlook and Teams
But: built for businesses, has no idea of students or modules, needs an administrator to set up

Calendly
Good: polished and easy to use
But: reminders cost about 10 US dollars per lecturer per month, and every lecturer has a separate link

Moodle Scheduler
Good: made for teaching
But: only works inside one course and needs an administrator to install it

THE GAP
None of them gives every CTU student one free place to book any lecturer.

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THE PLAN

Requirements and **how we worked**

WHAT IT MUST DO

9 functional requirements

- Log in as a student or a lecturer
- Lecturers publish, edit and remove their slots
- Students book, change or cancel with a reason
- No double bookings
- Confirmation and reminder emails, and booking history

7 non-functional requirements

- Quick to book and pages that load fast
- Works on a phone and a laptop
- Hashed passwords, and each user only sees their own data

HOW WE WORKED

Agile Scrum in one-week sprints

0. **Sprint 0 | 16 to 19 Oct** — Requirements, SRS and design
1. **Sprint 1 | 20 to 23 Oct** — Build the six features and first tests
2. **Sprint 2 | 24 to 29 Oct** — User testing, fixes and the final report

Each member worked on a GitHub branch, and pull requests were reviewed before merging into main.

CampusBook | SE2 Group 3 | CTU Training Solutions 7

THE SOLUTION

CampusBook: **six must-have features**

Two roles
Students and lecturers log in and each sees their own pages

Lecturer slots
Lecturers publish the times they are free each week

Book with a reason
Students book, change or cancel and say why they are coming

Email and reminder
A confirmation straight away and a reminder before the meeting

Booking history
Both roles see upcoming and past bookings

Phone and laptop
Every page works on a small phone screen and a laptop

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THE BUILD

How it is **built**

WHAT YOU SEE
React web app
The pages in the browser: log in, book a slot, my bookings, lecturer slots

→

THE RULES
Node.js and Express API
Checks who you are and what you may do, then applies the booking rules

→

THE DATA
MySQL database
Stores users, lecturers, slots, bookings and emails in five tables

Emails
Nodemailer sends the confirmation. A timer job checks every minute and sends reminders 60 minutes before.

Security
Passwords are stored scrambled (bcrypt). After logging in, each request carries a token that proves who you are.

THE SOLUTION

One booking, **step by step**

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6
Lecturer publishes a free time	**Student picks a lecturer and an open time**	**Student adds a reason and confirms**	**Both get a confirmation email**	**Reminder email before the meeting**	**Change or cancel opens the time again**

No messages back and forth. The moment a time is booked, no other student can take it.

THE BUILD

The hard part: **two students, one slot**

OUR TEST

8 requests
for the same slot at the same moment, from three students

1 booking
The other 7 saw "Sorry, that slot is no longer available"

The slot is locked while it is booked
The second request waits for the first to finish, then sees the slot is taken.

The database is the safety net
Even if the code had a bug, the database refuses a second booking for the same slot.

Lecturers cannot break a booking
A slot that is already booked cannot be edited or removed, and slots cannot overlap or be in the past.

THE BUILD

Testing found 10 problems. **We fixed all of them**

Two pages showed no times
The Lecturer slots and Book a slot pages used different names for the times than the API. We matched the names.

Dates moved by one day
A time-zone conversion shifted some dates. The database now sends the date as plain text.

No way to change a booking
My bookings only had Cancel. We added a Change time button that moves the booking.

Passwords could be guessed forever
After 10 wrong passwords in 15 minutes, the login now blocks more tries for a while.

***Lesson:** most bugs were at the joins between our parts, so agree on names and test the joins early.*

THE RESULTS

What the testing **showed**

USER TESTING
4 students and 2 lecturers

4 of 6
rated it easy to use

4.5 out of 5
students' average ease rating (lecturers 2.5)

9 of 9
functional requirements met

10 of 10
defects fixed, none left open

389 ms
slowest page load

1 of 8
booking requests for the same slot got through

Also passed: students cannot see each other's bookings, and every page works on a phone and a laptop.

THE RESULTS

Did we meet our **objectives?**

1	**Find out how consultations work now**	13 students answered the survey	Achieved
2	**Compare existing booking tools**	Bookings, Calendly and Moodle Scheduler	Achieved
3	**Build the six must-have features**	All 9 functional requirements met	Achieved
4	**Test it**	35 of 36 tests pass, pages under 0.4 s	Mostly achieved
5	**Evaluate it with real users**	4 of 6 found it easy to use	Partly achieved

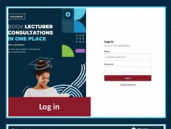
WHAT WE LEARNED

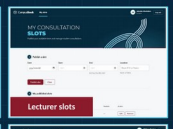
Five things **we learned**

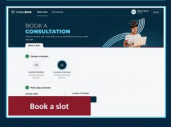
Start with the users	The survey, not our guesses, decided the features.
Agree on the details first	Most bugs came from parts using different names for the same data.
Put the rules on the server	Screens can be bypassed. The API and database cannot.
Test early and test the joins	Joining everyone's work is where the problems showed up.
Work in branches	Each member worked on a GitHub branch and merged with pull requests.

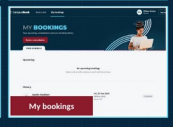
LIVE DEMO

Let us show **you CampusBook**





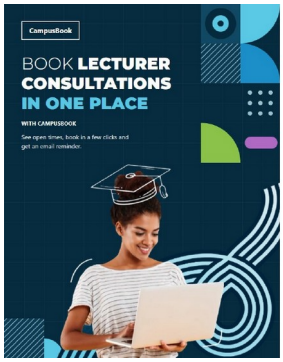




WHAT WE WILL SHOW

1. Lecturer logs in and publishes a free time
2. Student logs in, finds that time and books it with a reason
3. The confirmation email arrives and the time is gone for others
4. Student changes the time, then cancels
5. The slot is open again on the lecturer page

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Thank you

Questions?

CampusBook: book lecturer consultations in one place

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